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1. LIST OF ABBREVIATIONS

Abbreviation		Full Form/Description
%	:	Percentage
<	:	Less than
≥	:	Greater than or equal to
° C	:	Degree Celsius
µg	:	Microgram
µL	:	Microliter
a	:	anti
am	:	Ante meridiem
APSL	:	Aurigene Pharmaceutical Services Limited
ATCC	:	American Type Culture Collection
BW	:	Body Weight
BSA		Bovine serum albumin
BSB	:	Brilliant stain buffer
CFR	:	Code of Federal Regulations
Cm^2	:	Centimeter Square
CPCSEA	:	Committee for the Purpose of Control and Supervision of Experiments on Animals
FBS	:	Fetal Bovine Serum
FDA	:	Food and Drug Administration
g	:	Gram
GLP	:	Good Laboratory Practice
h	:	Hour
HEPES	:	4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid
HRP	:	Horseradish peroxidase
i.p	:	Intraperitoneal
IAEC	:	Institutional Animal Ethics Committee
IPA	:	Iso Propyl Alcohol
K2EDTA	:	Potassium Ethylene Diamine Tetra Acetic Acid
kg	:	Kilogram
LC-MS/MS	:	Liquid Chromatography-Tandem Mass Spectrometry
LPS	:	Lipopolysaccharide
Ltd	:	Limited

Abbreviation		Full Form/Description
mg	:	Milligram
min	:	minute
mL	:	Milliliter
mm	:	Millimeter
mM	:	Milli Molor
n / No	:	Number
NECA	:	5′-(N-Ethylcarboxamido) adenosine
NA	:	Not applicable
PH	:	Pharmacology
PGE2	:	Prostaglandin E2
PK	:	Pharmacokinetics
pm	:	Post meridiem
og	:	Oral gavage
PPE	:	Personal Protective Equipment
Tid	:	*ter in die*, thrice a day
TNF-α	:	Tumour Necrosis Factor- α
TV	:	Tumour Volume
RPMI-1640	:	Roswell Park Memorial Institute 1640 medium
RBC	:	Red blood cells
rpm	:	Revolutions per minute
SOP	:	Standard Operating Procedure
w/v	:	Weight / Volume

2. RESPONSIBILITIES / STUDY PERSONNEL

Study No.	:	[redacted]
Study Status	:	Non-GLP
Study Title	:	*In vivo* efficacy and T cell infiltration study of [redacted] in combination with anti-PD1antibody in CT26 syngeneic Tumor model
Study Director	:	Sandeep Mahankali. Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad-500 049, Telangana, India Tel no.: +91-40-44657522 E-mail: sandeepm@aurigeneservices.com
Head Function	:	D. Ravi Krishna Babu Department of Pharmacology, Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad-500 049, Telangana, India Tel no.: +91-40-44657345 E-mail: ravikrishna_b@aurigeneservices.com
Responsible Scientists	:	Ravi Teja, M. Pharm. Bhanu Teja, M. Pharm. Gouthami, M. Pharm. Prashant Yallappa Vadnal, M. Pharm. Ch. Sandeep Reddy, M. Pharm. Arundhuti Mallik, M.Sc.
Animal Welfare Officer	:	Dr. Sachin Bhalchandra Pagrut M.V. Sc.
Test Facility	:	Aurigene Pharmaceutical Services Limited Bollaram Road, Miyapur, Hyderabad-500 049, Telangana, India

		Telephone no.: +91-40-44657777 Fax no.: +91-40-44657438
Sponsor & Sponsor's Representative	:	[redacted]

3. PROPOSED STUDY SCHEDULE

Study initiation date (Date study director signs the study protocol)	:	13 DEC 2022
Cell culture initiation	:	08 DEC 2022
Acclimatization initiation	:	12 DEC 2022
Tumour cell inoculation dates	:	19 DEC 2022
Experimental completion date (In-life phase completion)	:	Animals will be sacrificed on day 6 and 15 of treatment initiation (exact time and date will be recorded in the report and Lab notebook)
Draft report (Excel format)	:	Three weeks from experiment completion
Study completion date (Date study director approves the final Excel / Power point report)	:	Four weeks from experiment completion date (exact date will be mentioned in the report)

The proposed study schedule will serve as a guide and the actual dates may be subject to change and will be documented in the raw data and final study report.

4. REGULATORY COMPLIANCE AND APPROVALS

4.1. The Guidelines

The design of this study is based on the study objective(s) and the overall development strategy for the test item and is not based on any specific guidelines.

4.2. Compliance

The non-clinical laboratory study described in this protocol is not intended to comply with the United States (US) Food and Drug Administration (FDA) Good Laboratory Practice (GLP) Regulations for non-clinical studies, 21 Code of Federal Regulations (CFR) Part 58.

This study will be conducted as per the mutually agreed study protocol, amendments (if any) with the sponsor and applicable SOPs of APSL.

4.3. Institutional Animal Ethics Committee (IAEC) Approval

This study will be conducted with the approved procedures of Institutional Animal Ethics Committee. Aurigene IAEC registration number is 812/PO/RcBi/S/04/CPCSEA and according to Aurigene Institutional Animal Ethics Committee approved protocol number APSL/IAEC/PH/116-05/03-2022.

5. SAFETY PRECAUTIONS

Gloves, cap, face mask, lab coats and/or other appropriate personal protective equipment (PPE) will be used to handle the test and /or standard compound to ensure adequate personal health and safety and to avoid inhalation and skin contact with the test item. In case of eye contact, the eyes must be washed thoroughly with water and medical treatment sought.

In case of skin contact, skin must be washed with soap and water with subsequent medical aid. Appropriate PPE will always be worn and care must be taken while thawing of frozen vials as conversion of liquid phase back to its gas phase may result in the vessel exploding.

Leak and Spillage Procedures: Area of spillage will be ventilated, appropriate PPE (gloves, cap, face mask, gowns and safety glasses) will be worn before entering the area. Spillage area including floor and walls will be flushed with water and wet mop or vacuum pump will be used to pick up spilled product. Unnecessary and unprotected personnel will be kept away from entering into the spillage area.

6. OBJECTIVE

The objective of the study is to evaluate efficacy of [redacted] in combination with anti-PD1 mAb in CT26-tumour bearing female BALB/c mice.

7. *IN VIVO* EFFICACY STUDY

7.1. Materials and Methods

7.1.1.Test item information

Name	:	[redacted]
Synonyms	:	[redacted]
Batch Number	:	[redacted]
Appearance	:	Off-white solid
Purity HPLC, 248nm)	:	>99%
Manufacture Date	:	08/03/2022
Retest Date	:	04/04/2023
Correction Factor	:	1.010
Storage Condition	:	Room temperature, protected from light
Relative Molecular Mass	:	[redacted]
Molecular formula	:	[redacted]

Name	:	InVivoPlus Anti-mouse PD-1 (CD279) control
Clone, isotype		RMP1-14, Rat IgG2a, κ
Catalogue	:	BP0146
Batch Number	:	825822J1
Concentration	:	9.78 mg/mL
Molecular weight		150 kDa
Storage Condition	:	2-8°C

Name	:	rat IgG2a isotype control
Clone		2A3
Catalogue	:	BP0089
Batch Number	:	815021S1
Concentration	:	9.35 mg/mL
Molecular weight		150 kDa
Storage Condition	:	2-8°C

7.1.2. Vehicle Information

- For [redacted] compound 90% PEG-400 + 10% solutol HS15 will be used as vehicle.

7.1.3. Route of Administration

- Test and vehicle formulations will be administered by oral gavage route thrice daily.
- Anti PD-1 mAb and isotype controls will be administered by intraperitoneal (*i.p*) route once every 3 days.

7.1.4. Dose Formulation – Preparation, Frequency of Preparation, Storage and Usage

Test item name, Strength/ Dose concentration	Formulation preparation, frequency & storage	Storage condition of test compounds/ items
[redacted] 25 mg/kg, 5 mL/kg (5 mg/mL)	[redacted] formulation stability for 25 mg/kg strength was established by Sponsor and is stable for 7 days. Formulation prepared will be stored at room temperature.	[redacted] stored at room temperature in the dark.
InVivo Plus anti-mouse PD-1 (CD279) Clone: RMP1-14, 10 mg/kg/10 mL (1 mg/mL)	On the day of injection, antibodies vials will be taken into laminar airflow unit for dilution preparations. 1 mg/ml of antibody formulations will be prepared by using InVivoPure pH 7.0 dilution buffer as in section 7.5.	InVivo Plus anti-mouse PD-1 (CD279), InVivo Plus rat IgG2a isotype control and buffers stored in refrigerator at 2-8°C.
InVivo Plus rat IgG2a isotype control, Clone: 2A3, 10 mg/kg/10 mL, (1 mg/mL)	On the day of injection, antibodies vials will be taken into laminar airflow unit for dilution preparations. 1 mg/ml of antibody formulations will be prepared by using InVivoPure pH 6.5 dilution buffer as in section 7.5.	

7.2. Test System Details

Species	:	Mice
Strain	:	BALB/c (BALB/cAnNTac)
Source	:	Vivo Bio Tech Limited, Hyderabad. Vivo Bio Tech is authorized breeder and distributor for Taconic mouse in India.
Justification for the selection of species/strain	:	CT26 syngeneic tumour model is established in BALB/C mice.
Total no. of mice	:	104
Total number of groups	:	4
No. of Mice / group	:	16
Sex	:	Female
Acclimatization period	:	Minimum 3 days
Age at the start of Acclimatization	:	7 to 8 weeks. Animals used in the study will be nulliparous and non-pregnant.
Body weight range	:	~17 to ~25 grams (at inoculation with CT26, exact range will be provided in the report).
Identification	:	After randomization to treatment group, all mice in a cage will be identified by tail marking with permanent marker throughout the experimental period. All the cages in the study will be identified by cage cards with details of study number, group, sex, animal number and study director.

7.3. Husbandry

7.3.1.Care and Housing

Housing system	:	Environmentally monitored air-conditioned room
Temperature and Humidity	:	22 ± 3 °C and 50 ± 20%, respectively
Lighting Cycle	:	12h dark /12h light cycle

Cages	:	Polycarbonate cage for mouse (Floor area 408 cm^2 and height 13 cm) with stainless steel top grill having facilities for holding pelleted food and drinking water bottle.
Housing	:	5 animals will be housed in each cage till randomization. After randomization 4 animals will be housed in each cage. In case of fighting or ≥ 10% body weight loss during study period the affected mice will be housed 1-3 per cage.
Bedding material	:	ATNT qCOB - Corn cob (gamma irradiated) and Mouse Tunnels (environmental enrichment).
Cage/bedding material change frequency	:	Weekly once

7.3.2. Diet and Water

Animals will be fed *ad libitum* with gamma irradiated rodent pellet diet (Type 1334 P - 14.9% protein rodent maintenance diet, Altromin Spezialfutter GmbH & Co. KG, ATNT Laboratories – (batch number will be provided in the report) throughout the experimental period. Reverse osmosis treated water will be provided *ad libitum* in polypropylene bottles filled twice weekly. The analysis reports of feed and water will be enclosed in the study file.

7.4. List of reagents/ kits

Name	Make	Catalog no.	Batch/ Lot no.	Expiry date
DPBS media	Sigma	59321C	SLCJ2830	31 Jan 2025
FBS	FBS	30-2020	80219215	Apr 2026
Trypsin	Gibco	25200-056	2342253	Sep 2023
RPMI 1640 Media	Gibco	A10491-01	2413330	May 2023
Pen Strep	Gibco	15140-122	2441841	30 Apr 2023
Trypan blue stain 0.4%	Invitrogen	T10282	2502913	NA
Freezing Medium	Gibco	12648-010	2450318	30 May 23
Countess chamber slides	Thermo fisher	C10283	I30B26132	NA
Solutol HS15	Sigma	42966	BCCF4839	
PEG 400	Sigma	06855	BCCB5243	
Tween 80	Sigma	P4780	K017022	Apr 2023
Water	Rankem	W0020	52210008815	26 Jul 2023
Tumour dissociation Kit	Miltenyi Biotech	130-096-730	RNBK5280	Oct 2023
Red Blood Cell Lysing Buffer Hybri-Max	Sigma	R7757	SLCJ2830	31 Jan 2025

Name	Make	Catalog no.	Batch/ Lot no.	Expiry date
96 Well TC-Treated Microplates	Corning Costar	3799	35621037	21-Apr-2024
CytoFlex Daily QC beads	Beckman Coulter	B53230	12APA	10 Jun 2025
FlowClean Cleaning Agent	Beckman Coulter	A64669	3991091F	13 Oct 2023
Sheath Solution	Beckman Coulter	B51503	6410099	29 Sep 2023
Stain Buffer	BD	554656	2063415	29 Mar 2024
Fixation buffer	BD	554655	335817	27 Jan 2023
Fc Block Ab	BD	553142	1137227	28 Feb 2029
Brilliant stain buffer	BD	566385	2062212	31 Aug 2023
InVivoPlus anti-mouse PD-1 (CD279)	2BScientific	BP0146-25MG	825822J1	24 Nov 2023
InVivoPlus rat IgG2a isotype control	2BScientific	BP0089-25MG	815021S1	24 Nov 2023
FoxP3 buffer set Fixation concentrate	BD Bioscience	560409	2006984	30 Nov 2023
FoxP3 Permeabalisation concentrate	BD Bioscience	560409	2028265	31 Jan 2023
FITC Rat Anti-Mouse CD4	BD Bioscience	553046	1116730	28 Feb 2027
PerCP Rat Anti-Mouse CD8a	BD Bioscience	553036	1117051	31 Mar 2024
PE Rat Anti-Mouse CD223	BD Bioscience	552380	1097461	31 Oct 2024
PE-Cy™7 Mouse Anti-Mouse NK-1.1	BD Bioscience	552878	1286186	30 Sep 2023
Alexa Fluor® 647 Rat anti-Mouse Foxp3	BD Bioscience	560401	2178603	30 Apr 2023
APC-Cy™7 Rat Anti-Mouse CD3 Molecular Complex	BD Bioscience	560590	2032893	31 Dec 2023
BV421 Rat Anti-Mouse CD25	BD Bioscience	564370	1131968	31 Aug 2023
FITC Mouse Anti-Mouse I-A[k]	BD Bioscience	553540	2010051	31 Oct 2023
PE Rat Anti-Mouse CD274 (PD-L1)	BD Bioscience	568085	1342601	02 Aug 2024
PE-Cy™7 Rat Anti-Mouse CD86	BD Bioscience	560582	2089240	31 Jul 2023
Alexa Fluor® 647 Rat Anti-Mouse CD73	BD Bioscience	561543	1292157	31 Jul 2026

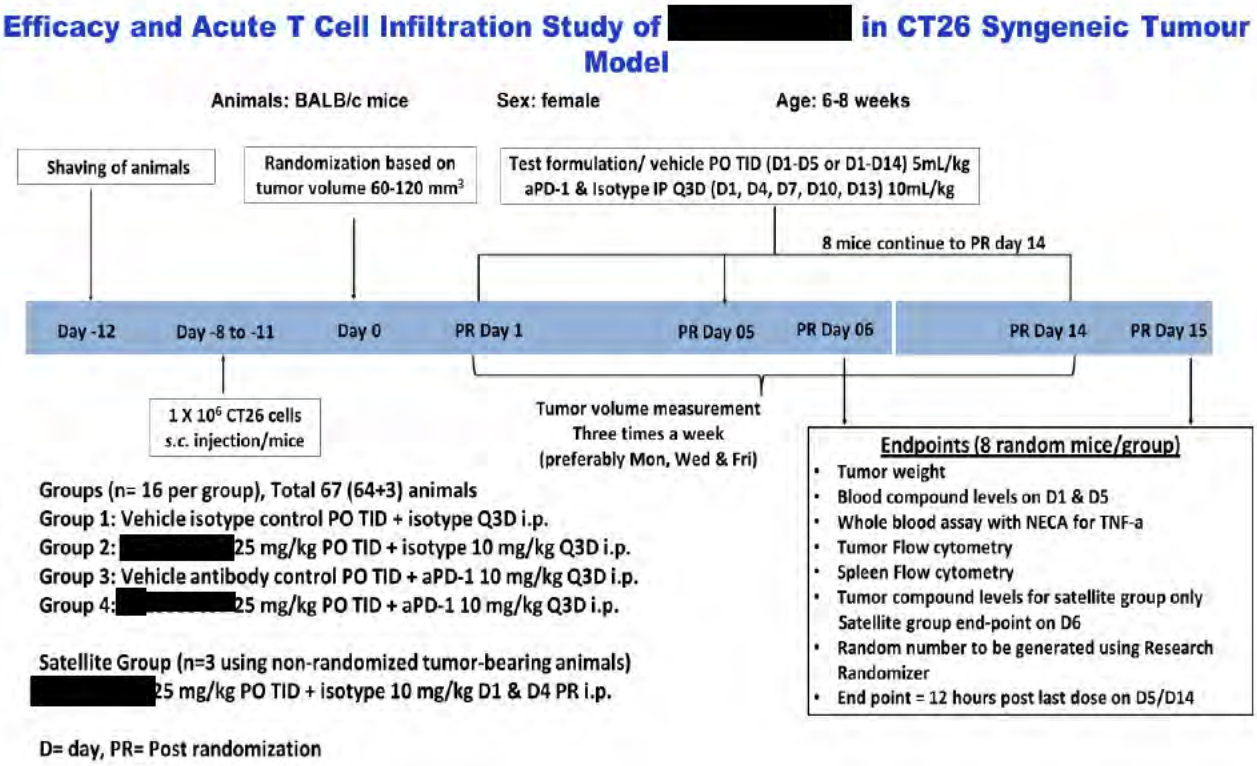
Name	Make	Catalog no.	Batch/ Lot no.	Expiry date
BV421 Rat Anti-Mouse F4/80	BD Bioscience	565411	2251487	31 Jan 2025
BV650 Rat Anti-Mouse CD45	BD Bioscience	563410	2087857	31 Jul 2023
PerCP anti-mouse CD11c Antibody	BioLegend	117326	B288262	NA
Fixable Viability Stain 510	BD Bioscience	564406	1153697	20 Jun 2023
BSA	Sigma	A9418	SLBZ6149	Sep 2023
Cell strainers	Mitenyi	130-098-462	5220420004	19 Apr 2025
Gentle Macs tubes	Miltenyi	130-093-237	52200801001	15 Sep 2025
2-Mercaptoethanol (14.3 M)	Sigma	M3148	BCBV8182	Apr 2023
DMSO	Sigma	D2650	RNBK7771	May 2024
LPS	Sigma	L2880	0000135216	Aug 2027
PGE2	Sigma	P0409	SLCG3303	TBU
HBSS	Sigma	H8264	SLCK3918	Aug 2023
TNF-α ELISA kit	RD systems	DY410	P260637	24 Feb 2025
DuoSet ELISA Ancillary Reagent Kit 2	RD systems	DY008	P262743	20 Jul 2023
MEM non-essential amino acids	Gibco	11140050	2517549	28 Feb 2024
Heparin sodium injection (5000IU/mL)	Biological E Limited	SL-MB/2013-105 (ML NO)	NLB042	Feb 2024

7.5. Equipment to be used

Equipment name	Make & Model number
Biosafety cabinet	Thermo scientific Esco class II
Biosafety cabinet	BioKlenz
Digital Caliper	Fowler sylvac
Precision balance	Sartorius BSA3202S-CW
Micro weighing Balance	Sartorius SE2
Eppendorf centrifuge	Eppendorf 5417R
Anesthesia Machine	Matrx M 4000
pH Meter	EUTECH Instruments PC 700
Homogenizer	Remi Motors RQ-12 7A
Sonicator	Bransonics 3210E-MTH

Octo Dissociator with heater	Miltenyi Biotech, gentle MACS
Flow cytometer	Beckman Coulter, cytoFLEX
ELISA plate reader	Biotek, Power wave XS

7.6. Study Design



7.7. Cell line propagation and inoculation

Preparation of media for cell culture:

CT26 cancer cell line procured from ATCC (CRL-2638), revived, cryopreserved previously, and stored in liquid nitrogen at APSL-Hyderabad cell line repository will be used for this experiment. CT26 cell line will be grown in RPMI-1640 media (contains 10 mM HEPES, 1 mM sodium pyruvate, 4.5 g/L Glucose, 1500 mg/L sodium bicarbonate) supplemented with 10% FBS and 1% penicillin streptomycin. Frozen vial of CT26 cells will be revived in T-flask containing complete RPMI-1640 media and incubated at 37±1°C in an incubator supplied with 5% carbon dioxide. Cell expansion will be carried out in T-175cm^2 flasks. After the final step of splitting, cells will be processed for injection when the cells reach exponential phase. Cells will be trypsinized with 1x Trypsin-EDTA solution and cell suspension from T-175cm^2 flask will be transferred to 50mL tubes and centrifuged

at 1200 rpm for 5 minutes at controlled room temperature to obtain cell pellet. Supernatant will be discarded, and cell pellets will be re-suspended in media and counted using hemocytometer. Cell pellet will be re-suspended in RPMI-1640 plain media at a final concentration of $10x10^6$ cells/mL.

Tumour Implantation: On day -12 right flank region of the animals will be shaved using animal clipper under anesthesia. On day -11 (to -8) tentatively after measurement of body weights, under 3% isoflurane with oxygen, animals right flank area will be sterilized with IPA, and 100 µl cell suspension containing $1x10^6$ cells will be injected slowly (10 million cells per ml) with a 1mL syringe into subcutaneous region, avoiding leakage and making round bulge of injection. After injection, animals will be put back in their home cage and observed until recovery.

7.8. Tumour Volume Measurement

7-11 days after tumour implantation, the tumour volumes will be measured using digital Vernier calipers in two perpendicular planes and the tumour volume will be calculated using the formula:

$$\text{Tumour volume (TV)} = (D \times d^2)/2$$

Where D represents the "largest tumour diameter (in mm)"; d represents the "smallest tumour diameter (in mm)".

7.9. Grouping and Allocation of Animals

When the tumour volume is 60-120 mm^3 (or 50 to 100 mm^3, ~2-fold range of desired selection) tumour bearing animals will be randomized into 4 groups with 16 animals in each group, in one cohort, as mentioned below in the table

. Efficacy Study Groups					
Group No.	**Dosage from D1 PR**			**No. of Mice**	**Animal Numbers**
	Oral treatment	**Dosing period**	**Dose Volume (mL/kg)**		
G1	Vehicle control + Isotype	TID, o.g + Q3D, i.p.	5 mL/kg for oral dosing and 10 mL/kg for antibody dosing	16	1-16
G2	[redacted] 25 mg/kg/5mL + Isotype 10mg/kg/10mL	TID, o.g + Q3D, i.p.		16	17-32
G3	Vehicle control + aPD-1 10 mg/kg/10mL	TID, o.g + Q3D, i.p.		16	33-48
G4	[redacted] 25 mg/kg/5mL + aPD-1 10 mg/kg/10mL	TID, o.g + Q3D, i.p.		16	49-64

Note: Time of the study endpoint (day 6 and day 15), including tumour collection, will be approximately 12 h after day 5 or day 14 dosing. Exact time for each mouse will be recorded in the lab AM dosing and PM dosing will be at approximately 6:00 AM, 2:00 PM and 10:00 PM ± 30 minutes. Dosing times will be recorded on the dosing sheets; o.g -oral gavage, i.p -Intraperitoneal.

Any remaining tumour-bearing animals not randomized to either the efficacy or satellite group will be used for FACS trails, blank tumour, blank blood and plasma.

Remaining animals not randomized to any treatment group in main efficacy study, among them 3 mice will be randomized to a satellite group study.

Satellite Study Group					
Group No.	**Dosage from D1 PR**			**No. of Mice #**	**Animal Numbers**
	Oral treatment	**Dosing period**	**Dose Volume (mL/kg)**		
S1	[redacted] 25mg/kg + Isotype control 10 mg/kg	TID x 5 days, o.g + D1 and D4 i.p	10	3	65-67

7.10. Preparation of test formulations and Solutions

Note: Preparation procedure of reagents, vehicle and formulations is described for fixed volumes and concentrations. However, volume of each reagent, vehicle, formulations will be prepared as per experimental requirement

Preparation of 90% PEG-400 and 10% Solutol HS15 Vehicle:

90 mL of PEG-400 will be added to 10 mL of Solutol HS15

Formulation preparations:

[redacted] **5 mg/mL (For 12 ml):** Add 10.8 mL of PEG400 will be added to tube containing 60 mg of [redacted] compound. The tube will be vortexed for 1 minute and sonicated for 5 minutes, stirred for 10 minutes at 65ºC. Then 1.2 ml of Solutol HS15 will be added into tube, vortexed for 1 minute, sonicated for 15 minutes and stirred for 80 minutes at 65ºC. Formulation will be stored at 2-8ºC or room temperature and brought room temperature prior to dose administration. Compound is stable in this vehicle for 7 days at room temperature or 4ºC.

InVivo Plus anti-mouse PD-1 antibody (CD279) Clone: RMP1-14, 10mg/kg/10mL:

869 µL of anti-mouse PD1 antibody will be diluted with 7631 µL of InVivoPure pH 7.0 Dilution Buffer to make 8.5 ml of formulation with 1mg/mL concentration. Prepared formulations will be administered to all the animals in sterile condition.

InVivo Plus rat IgG2a isotype control, Clone: 2A3, 10mg/kg, 10ml/kg/10mL:

909 µL of rat IgG2a isotype control will be diluted with 7591 µL of InVivoPure pH 6.5 Dilution Buffer to make 8.5 ml of formulation with 1mg/mL concentration. prepared formulations will be administered to all the animals in sterile condition.

7.11. Treatment

For efficacy study, on day 0 (randomization day), animals will be randomized based on tumour volumes into 4 groups of equal number of animals in each group. Oral treatment will be administered by gavage as depicted in table efficacy groups (section 7.9) starting from day 1 and continued till day 5/ day 14 (G1-G4). The dose administration volume (5mL/kg for oral dosing & 10 mL/kg for intraperitoneal, body weight) will be calculated from the individual mouse body weight (BW) measured prior to dose administration. Body weight will be measured once daily before 2:00 AM dosing's, and this weight will be used for all oral dosing administrations till next day 6:00 AM dosing.

7.12. Study procedures

Post randomization treatment will be proceeded as mentioned in above section 7.11. On day 1 and day 5 post 8h of first dose, all animals will be bled (refer section 7.14 for blood collection procedure) for compound analysis.

On day 5 post dose 8h of bleeding, animals will be grouped to Set1 and Set2. 8 animals from each group (G1- G4) will be selected based on rank order of TV such that both sets will have similar tumor volume. Set 1 animals will proceed for day 6 termination whereas set 2 animals will be continued up to day14 dosing

Ranking order: Tumor volume will be sorted smallest to largest value of each group separately and ranking will be allotted.

Group No.	Treatment Groups	Rank order based on TV of day 5 or latest	
		Set 1 (Day 6 termination); n=8	Set 2 (Day 15 termination); n=8
G1	Vehicle control + Isotype	1,3,5,7,9,11,13,15	2,4,6,8,10,12,14,16
G2	[redacted] 25 mg/kg/5mL + Isotype 10mg/kg/10mL	17,19,21,23,25,27,29,31	18,20,22,24,26,28,30,32

Group No.	Treatment Groups	Rank order based on TV of day 5 or latest	
		Set 1 (Day 6 termination); n=8	Set 2 (Day 15 termination); n=8
G3	Vehicle control + aPD-1 10 mg/kg/10mL	33,35,37,39,41,43,45,47	34,36,38,40,42,44,46,48
G4	[redacted] 25 mg/kg/5mL + aPD-1 10 mg/kg/10mL	49,51,53,55,57,59,61,63	50,52,54,56,58,60,62,64

Note: Rank order will be allotted to minimize the TV variation in between the sets.

On day 6 post 12 h of last dose of day 5, the 8 animals previously selected (set 1) from efficacy treatment groups (G1-G4) and satellite group blood will be withdrawn for compound analysis and whole blood assay (refer section 8) (refer section 7.14 for blood collection procedure) and sacrificed under 3-5% isoflurane, tumors and spleens from animals in all treatment groups of set1 animals will be collected, weighed, and used for flow cytometer staining procedure (refer section 9). From the satellite study, tumors from all animals will be collected, wrapped in aluminum foil, flash froze in liquid nitrogen and stored at -60°C to -80°C for compound analysis by LCMS to assess pharmacokinetics.

The remaining 8 animals (set 2) will be continued with respective treatments till day14. On day 15, set 2 animals from all groups similar like day 6 procedure will be proceeded terminal collection post 12 h of last dose of day14 (refer section 7.14).

If it is necessary to remove an animal from study due to welfare concerns, then the day 15 endpoint will be performed as part of the culling process.

7.13. Observations

7.13.1. Mortality/Moribundity

Animals will be observed for mortality/morbidity twice daily throughout the experimental period.

7.13.2. Clinical Signs

Animals will be observed for clinical signs thrice daily (AM and PM) throughout the treatment period. Any animal showing abnormal clinical signs will be noted and will be closely monitored. Sponsor's representative will be notified promptly by email if clinical signs are observed.

7.13.3. Body Weight

Individual animal body weights will be recorded daily before the 2:00 PM (this BW will be considered till next day morning dosing) dosing administration of the test item and throughout the experimental period. If an animal is found moribund, with severe clinical signs or more than 20% drop in body weight from its highest body weight, it will be sacrificed humanely. This decision will be taken by study director in consultation with veterinarian and flagged with the Sponsor's representative.

% Body weight change will be calculated as: [(B weight $^{\text{Final day}}$ - B weight $^{\text{Initial day}}$)/ (B weight $^{\text{Initial day}}$)] *100

Initial day = day of randomization to study groups (day 0)

7.13.4. Tumour volume and weight measurements

Tumour volumes will be measured in all treatment group animals thrice a week (Monday, Wednesday, and Friday) using a calibrated set of digital Vernier calipers, as described in section 7.8. From an ethical viewpoint, any mouse with a tumour volume exceeding 3000 mm^3 will be humanely sacrificed and the study day 6 and 15 endpoints performed; any animal removal decision will be taken by the Study Director.

As a measure of efficacy % TGI (% Tumour growth inhibition) values will be calculated (based on availability of control animals for calculation purposes).

On day 15, 12 hours after day 14 last oral dosing, all remaining study animals will be sacrificed and blood, tumour tissues will be collected, and tumour weight will be recorded using weighing balance.

7.14. Study termination including blood and tumour sampling

On day 6 and day 15, at approximately 12 hours after the last dose (day 5 and day 14 respectively), the efficacy groups and satellite groups will be proceeded for termination.

Blood Collection:

Each animal will be anaesthetized under 3-5% isoflurane with oxygen for approximately 3-5 minutes before blood collection (40 µL and 350 µL for a total of 390 µL).

For Compound analysis: The 40 µL blood aliquot will be collected into a labeled 0.5 mL micro centrifuge tube added with 4 µL of 5% anticoagulant (K2 EDTA), gently agitated to mix the blood and EDTA. 30 µL will be taken into a new 0.5 mL centrifuge tube and 30 µL water added to lyse the cells and samples will be stored at -60°C to -80°C for future compound analysis by LCMS.

8.3. Preparation of tissue culture media for whole blood stimulation

1000 mL of assay buffer (RPMI buffer) will be prepared aseptically using 979 mL RPMI-1640, 10 mL MEM non-essential amino acid, 10 mL of 100X antibiotic Pen-Strep solution and 1 mL of 55 mM 2-mercaptoethanol.

8.4. Preparation of LPS stock

LPS 1mg/mL

Dissolve one vial containing 10 mg LPS in 2mL of RPMI buffer, aliquoted 100 µL per vial and stored at -70 ± 10°C to 5 mg/mL.

Under sterile conditions, one vial of 100 µL LPS aliquot will be allowed to thaw to room temperature and diluted to 500 µL to get a concentration of 1 mg/mL (1000 µg/mL) using RPMI-1640 medium and further dilutions will be carried out as mentioned in below.

LPS 10 µg/mL

Under sterile conditions, one vial of 100 µL LPS (1 mg/mL) will be diluted to 10 mL to get a concentration of 10 µg/mL using RPMI-1640 buffer. 10 µL of this solution will be added to respective wells to and volume will be made up to 100µL to get a final concentration of 1µg/mL.

LPS vehicle: RPMI -1640 buffer

8.5. Preparation of NECA stock 10 mM

Under sterile conditions, 648 µL DMSO will be added to 2 mg NECA to get concentration of 10 mM. Aliquots of 40 µL will be prepared and stored at 4°C until further usage.

Preparation of NECA stock 10 µM

10 µL of 10 mM solution will be diluted to 1 mL using RPMI buffer to get a final concentration of 0.1 mM (100 µM), from this solution 400 µL of solution will be diluted to 4 mL using RPMI buffer to get a final concentration of 10 µM. 10 µL of this solution will be added to respective wells to and volume will be made up to 100 µL to get a final concentration of 1000 nM (1 µM).

Preparation of NECA stock 1 µM

400µL of NECA 10µM solution will be diluted to 4mL using RPMI buffer to get a final concentration of 1µM. 10µL of this solution will be added to respective wells to and volume will be made up to 100µL to get a final concentration of 100nM (0.1 µM).

8.6. Preparation of PGE2 stock

Preparation of PGE2 1µM (100nM)

1 mM of PGE2 is prepared by dissolving one vial containing 1 mg PGE2 in 2.8 mL of DMSO, aliquoted 50 µL per vial and stored at -70 ± 10ºC.

Under sterile conditions, one vial of PGE2 will be allowed thaw to room temperature and 10µL of 1mM solution will be diluted to 1mL RPMI buffer to get a concentration of 10µM, from this solution 400µL of solution will be diluted to 4mL using RPMI buffer to get a final concentration of 1µM. 10µL of this solution will be added to respective wells to and volume will be made up to 100µL to get a final concentration of 100nM.

NECA/PGE2 vehicle: 4 µL of DMSO is added to 3996 µL of RPMI buffer. This vehicle will be used as vehicle control group and final DMSO concentration is 0.01%.

8.7. Whole blood collection procedure

8.7.1 Each animal will be anaesthetized using isoflurane anesthesia.

8.7.1 ~350 to 400 µL of blood will be collected into a centrifuge tube containing heparin as anticoagulant (5 µL of 5000 IU/mL).

8.7.1 Blood samples will be kept independently and not pooled between animals.

8.8. Incubation of whole blood

8.8.1. 20 µL of blood withdrawn will be added to 96 well plate in duplicates, stimulated with 1µg/mL LPS in combination with 1 and 0.1 µM NECA or 100 nM PGE2 (as negative control).

8.8.2. After 4 hours incubation at 37°C, centrifuge the whole blood assay plate at 2000 g for 10 minutes at ~4°C and collect supernatant. The supernatant will be frozen at -20°C until ready to be profiled by TNF-a ELISA.

8.9. Cytokine estimation

8.9.1. Preparation of reagents for ELISA (DuoSet ELISA Ancillary Reagent Kit 2)

8.9.1.1. Wash buffer: To prepare 1X wash buffer 20 mL of 25X wash buffer concentrate will be added to 480 mL Milli-Q water.

8.9.1.2. Reagent Diluent: To prepare 1% BSA reagent diluent 2 mL of reagent diluent concentrate 2 (10X) will be added to 18 mL Milli-Q water.

8.9.1.3. Substrate solution: Color Reagents A and B should be mixed in equal volumes within 15 minutes of use.

8.9.2. Procedure for estimation of cytokines (TNF-α) (DuoSet ELISA)

8.9.2.1. Preparation of Goat Anti-Mouse TNF-α Capture Antibody working concentrations: 48 µL of 100 µg/mL of stock vial will be diluted to 6 mL using carrier protein free PBS to get a concentration of 800 ng/mL.

8.9.2.2. Recombinant Mouse TNF-α Standard: 2000 pg/mL of standard will be prepared by diluting 10 µL of 200 ng/mL (Stored aliquot) to 1 mL using reagent diluents. A nine-point standard curve using 2-fold serial dilutions.

8.9.2.3. Biotinylated Goat Anti-Mouse TNF-α Detection Antibody working concentration: 100 µL of 4.5 µg/mL of detection antibody will be diluted to 6 mL using reagent diluents to get a working concentration of 75 ng/mL.

8.9.2.4. Coat the high binding 96 well plate by adding 50 µL of working concentrations of capture antibody (TNF-α) and incubate at 2-8^{0}C overnight.

8.9.2.5. Aspirate each well and wash with wash buffer, repeating the process two times for a total of three washes. Wash by filling each well with wash buffer (300 µL). After the last wash, remove any remaining wash buffer and blotting it against clean paper towels.

8.9.2.6. Block plates by adding 150 µL of reagent diluent to each well.

8.9.2.7. Incubate at room temperature for 1 hour.

8.9.2.8. Repeat the wash as described earlier.

8.9.2.9. Add 50 µL of undiluted sample or standards per well. Cover with an adhesive strip and incubate 2 hours at room temperature.

8.9.2.10. Repeat the washing procedure.

8.9.2.11. Add 50 µL of the detection antibody, diluted in reagent diluent, to each well. Cover with a new adhesive strip and incubate 2 hours at room temperature

8.9.2.12. Repeat the washing procedure.

8.9.2.13. Streptavidin-HRP: Each vial contains 2.0 mL of streptavidin conjugated to horseradish-peroxidase. Dilute to the working concentration (40-fold dilution) using reagent diluent.

8.9.2.14. Add 50 µL of the working dilution of streptavidin-HRP to each well. Cover the plate and incubate for 20 minutes at room temperature.

8.9.2.15. Repeat the washing procedure.

8.9.2.16. Add 50 µL of substrate solution to each well.

8.9.2.17. Incubate at room temperature until color develops. Avoid placing the plate in direct light.

8.9.2.18. Add 50 µL of stop solution to each well. Gently tap the plate to ensure thorough mixing.

8.9.2.19. Determine the optical density of each well immediately, using a microplate reader set to 450 nm and 570 nm for OD correction. Duplicate values will be averaged, and standard curve will be generated by four parameters logistic using Gen 5 software.

8.9.2.20. If any wells are above the limit of quantitation they will be diluted and re-assayed.

Note: Preparation procedure of reagents, is described for fixed volumes and concentrations. However, volume of each reagent will be prepared as per experimental requirement

8.9.3. Data analysis

For each animal in whole blood assay, TNF-α values observed in LPS with NECA/PGE2 will express as percentage of LPS alone stimulation. Whole blood assay (TNF alpha estimation; ng/mL) data will be calculated as Mean ± SD/SEM for each group using Microsoft Excel. Raw data will be entered into graph pad prism and significance will be calculated compared to vehicle control as mentioned in Section 11.2

9. TUMOUR AND SPLENOCYTE FLOW CYTOMETRY

9.1. Reagents/ Buffer Preparations

9.1.1. Preparation of Enzyme Mixture for Tumour samples

Prepare enzyme mix by adding 2.35 mL of RPMI 1640, 100 µL of Enzyme D, 50 µL of Enzyme R, and 12.5 µL of Enzyme A, from the Tumour dissociation Kit. Add 2.5 ml of enzyme mixture for each tumour weighing between 0.04 -1 g weights.

Calculate required enzyme volume according to tumour samples and weight.

For 32 tumors prepare for 38 reactions, prepare enzyme mix by adding 89.3 mL of RPMI 1640, 3800 µL of Enzyme D, 1900 µL of Enzyme R, and 475 µL of Enzyme A. Add 2.5 ml of enzyme mixture for each of tumour weighing between 0.04 -1 g weights.

9.1.1.1. Preparation of Enzyme D

Prepare Enzyme D by reconstitution of the lyophilized powder in each vial with 3 mL of RPMI 1640. Remaining solution can be stored as aliquots at –20°C. This solution is stable for 6 months after reconstitution. Fresh vial or already aliquoted vials can be used in this study.

9.1.1.2. Preparation of Enzyme R

Prepare Enzyme R by reconstitution of the lyophilized powder in the vial with 2.7 mL RPMI 1640.

Remaining solution can be stored as aliquots at –20°C. This solution is stable for 6 months after reconstitution. Fresh vial or already aliquoted vials can be used in this study.

9.1.1.3. Preparation of Enzyme A

Prepare Enzyme A by reconstitution of the lyophilized powder in the vial with 1 mL of Buffer A supplied with the kit. Do not vortex. Remaining solution can be stored as aliquots at –20°C. This solution is stable for 6 months after reconstitution. Fresh vial or already aliquoted vials can be used in this study.

9.1.2. Preparation of 55 mM 2-mercaptoethanol

3.85 µL 2-mercaptoethanol (14.3 M) will be added aseptically to 1 mL RPMI-1640 to obtain 55 mM 2-mercaptoethanol.

9.1.3. Preparation of wash buffer for tumour/ spleen collection and processing

500 mL of wash media will be prepared aseptically using 489.75 mL RPMI-1640, 10 mL FBS and 0.25 mL of 55 mM 2-mercaptoethanol.

9.1.4. Preparation of Fc block

Add 30 µL of Fc block solution diluted with 2970 µL of stain buffer. Fc block (CD16/32) will be used 50 µL per well (1:10 dilution).

9.1.5. Preparation of Antibodies

All antibodies should be used 50 µL per well after final dilution. Preparation captured in 9.3 table. For 32 tumour samples prepare for 3000 µL, 2000 µL containing 40 µL of each of all antibodies in 1680 µL of stain buffer.

9.1.6. Live/dead stain Preparation

9.1.6.1. Stock preparation

Bring FVS510 dye powder and 260 µL of fresh cell culture-grade Dimethyl Sulfoxide (DMSO, e.g., Sigma D2650) to room temperature. Add 260 µL of DMSO and vortex solution well. Inspect the solution and repeat vortex until the stock dye has fully dissolved. This is the stock solution for live/dead stain.

9.1.6.2. Storage

Upon arrival, store the dry dye desiccated and protected from light at -80°C until use. After reconstitution with DMSO, store the stock solution at ~-20°C in small aliquots. Do not use reconstituted dye after 90 days of storage. Discard the dye solution after 90 days post reconstitution with DMSO.

9.2. Staining Marker

Fluorochrome	Panel 1		Panel 2		Laser
	Antibody	BD cat #	Antibody	BD cat #	
FITC	CD4	553046	MHCII I-A[k]	553540	Blue
PerCP	CD8	553036	CD11c	117326	Blue
PE	Lag-3 CD223	552380	PD-L1	568085	Yellow
PE-Cy7	NK1.1	552878	CD86	560582	Yellow
APC/ Alexa 647	Foxp3	560401	CD73	561543	Red
APC-H7/cy7	CD3	560590	CD3	560590	Red
BV421	CD25	564370	F4/80	565411	Violet
BV510	FVS510	564406	FVS510	564406	Violet
BV650	-		CD45	563410	Violet

9.3. FACS – FMO Control and Samples staining buffer composition and preparation

Panel 1			
Control/ sample name	**N=**	**Composition**	**Preparation of Antibody mixture**
Unstained Tumour/Spleen control	7	50 µL Stain buffer	Stain buffer
FVS510 FMO Control	2	FVS510 alone	150 µL containing 3 µL of FVS510 in 147 µL of 1xBSB
CD4 FMO Control	2	50 µL all antibody stain except CD4	150 µL containing 3 µL each of all antibodies (except CD4) in 129 µL of 1xBSB
CD8 FMO Control	2	50 µL all antibody stain except CD8	150 µL containing 3 µL each of all antibodies (except CD8) in 129 µL of 1xBSB
LAG-3 CD223 FMO Control	2	50 µL all antibody stain except Lag-3 CD223	150 µL containing 3 µL each of all antibodies (except Lag-3 CD223) in 129 µL of 1x BSB

Panel 1			
Control/ sample name	**N=**	**Composition**	**Preparation of Antibody mixture**
NK1.1 FMO Control	2	50 µL all antibody stain except NK1.1	150 µL containing 3 µL each of all antibodies (except NK1.1) in 129 µL of 1xBSB
Foxp3 FMO Control	2	50 µL all antibody stain except Foxp3	150 µL containing 3 µL each of all antibodies (except Foxp3) in 129 µL of 1xBSB
CD3 FMO Control	2	50 µL all antibody stain except CD3	150 µL containing 3 µL each of all antibodies (except CD3) in 129 µL of 1xBSB
CD25 FMO Control	2	50 µL all antibody stain except CD25	150 µL containing 3 µL each of all antibodies (except CD25) in 129 µL of 1xBSB
FVS510 FMO Control	2	50 µL all antibody stain except FVS510	150 µL containing 3 µL each of all antibodies (except FVS510) in 129 µL of 1xBSB
Stained Tumour/ Spleen control	(32 samples)	50 µL all antibody stain	2000 µL containing 40 µL each of all antibodies (40*8=320) in 1680 µL of 1xBSB

Panel 2			
Control name	**N=**	**Control composition**	**Preparation of Antibody mixture**
Unstained Tumour/spleen control	7	50 µL Stain buffer	1x Brilliant Stain buffer
FVS510 FMO Control	2	FVS510 alone	150 µL containing 3 µL of FVS510 in 147 µL of 1x BSB
MHCII I-A[k] FMO Control	2	50 µL all antibody stain except MHCII I-A[k]	150 µL containing 3 µL each of all antibodies (except MHCII I-A[k]) in 129 µL of 1x BSB
CD11c FMO Control	2	50 µL all antibody stain except CD11c	150 µL containing 3 µL each of all antibodies (except CD11c) in 129 µL of 1x BSB
PD-L1 FMO Control	2	50 µL all antibody stain except PD-L1	150 µL containing 3 µL each of all antibodies (except PD-L1) in 129 µL of 1x BSB
CD86 FMO Control	2	50 µL all antibody stain except CD86	150 µL containing 3 µL each of all antibodies (except CD86) in 129 µL of 1x BSB

Panel 2			
Control name	**N=**	**Control composition**	**Preparation of Antibody mixture**
CD73 FMO Control	2	50 µL all antibody stain except CD73	150 µL containing 3 µL each of all antibodies (except CD73) in 129 µL of 1x BSB
CD3 FMO Control	2	50 µL all antibody stain except CD3	150 µL containing 3 µL each of all antibodies (except CD3) in 129 µL of 1x BSB
F4/80 FMO Control	2	50 µL all antibody stain except F4/80	150 µL containing 3 µL each of all antibodies (except F4/80) in 129 µL of 1x BSB
FVS510 FMO Control	2	50 µL all antibody stain except FVS510	150 µL containing 3 µL each of all antibodies (except FVS510) in 129 µL of 1x BSB
CD45 FMO Control	2	50 µL all antibody stain except CD45	150 µL containing 3 µL each of all antibodies (except CD45) in 129 µL of 1x BSB
Stained Tumour/Spleen control	(32 samples)	50 µL all antibody stain	2000 µL containing 40 µL each of all antibodies (40*9=360) in 1640 µL of 1x BSB

Note: All the above antibodies will be prepared in 1X brilliant stain buffer (BSB)

9.4. Tumour cell collection

9.4.1 Collected tumor and spleen samples will be chopped into small pieces of 1-3 mm^3 and transferred to gentle MACS C tubes containing 2.5 ml of enzyme mixture.

9.4.2 Tumour/ spleen samples will be kept independently and not pooled between animals. However, for FMO control samples these will be pooled.

9.4.3 Tightly close C Tube and attach it upside down onto the sleeve of the gentle MACS dissociator.

9.4.4 Run the gentle Protocol 37C_m_TDK_1 for MACS CT-26 tumors and spleen m-spleen-01-01 and will run for 3 times

9.4.5 Add 1 mL of FBS and 6.5 ml of RPMI media to neutralize enzyme mixture.

9.4.6 Cell suspension is collected by using 70 µm cell strainer in 50 ml centrifuge tube and squash tumors by pressing several times with the rubber end of a 10 mL syringe plunger. If required add media to ensure complete cell suspension is strained through cell strainer.

9.4.7 Centrifuge the collected cell suspension at 450 g at 2-8ºC for 10 minutes and wash the cell pellet twice using cold wash buffer.

9.4.8 Add 1 ml RBC lysis solution to above pellet. Keep centrifuge tubes in ice for 1-2 minutes to lyse RBCs.

9.4.9 Add 10 mL of cold wash buffer and centrifuge at 450 g at 2-8°C and wash twice with cold wash buffer.

9.4.10 Obtained pellet will be re-suspended in 0.5 mL of wash buffer and cells will be counted using cell counter with trypan blue solution.

9.4.11 Cell number will be adjusted to ~10 x10^6/mL with RPMI media and will be kept on ice until ready to plate into 96 wells plate. 100 µL of cell suspension will be added to each well to obtain 1x10^6 cells per well.

9.4.12 Add 1 million corresponding sample cells to each well in 96 well plate and centrifuge at 450 g at 2-8°C for 5 minutes.

9.5. Blocking

Discard supernatant and re-suspend the cells in 50 µL of diluted Fc block antibody to each well and incubate for 10 minutes at room temperature.

9.6. Fixable Viability Stain 510 labeling of cells and antibody staining

Panel -1

9.6.1 Prepare cells for flow cytometry staining using sodium azide-free buffers.

9.6.2 Wash cells one time in Dulbecco's Phosphate Buffered Saline (1X DPBS).

9.6.3 Resuspend cells at 1x10^6 cells/ml in 1X DPBS.

9.6.4 Add 1 µL of the BD Horizon Fixable Viability Stain 510 Stock Solution for each 1million cell suspension (1:1000) and vortex immediately. If required, titrate the dye for optimal performance.

9.6.5 Incubate the mixture for 15 minutes at room temperature protected from light.

9.6.6 Wash cells twice with stain buffer.

9.6.7 Decant the supernatant and gently mix to disrupt the cell pellet.

9.6.8 Stain, fix and permeabilize cells as required.

9.6.9 Discard supernatant and re-suspend the cells in 50 µL of diluted Fc block antibody to each well and incubate for 15 minutes at room temperature.

9.6.10 Surface stain antibodies (except Foxp3 intracellular markers) are added 50 µL per well after final dilution on top of Fc blocked wells and incubated for 30 minutes at RT in the dark.

9.6.11 Fix the cells, gently re-suspend pellet in residual volume of staining buffer and then add 100 µL of freshly prepared cold 1×BD Pharmingen Mouse Foxp3 Fixation Buffer. Mix well. Incubate for 30 minutes at 4°C in the dark.

9.6.12 Centrifuge 500 g for 5 minutes and remove fixative.

9.6.13 Wash cells and re-suspend each pellet in 100 µL of freshly prepared pre-warmed 1×BD Pharmingen Mouse Foxp3 Permeabilization buffer, and centrifuge 500 g for 5 minutes. Remove permeabilization buffer.

9.6.14 To permeabilize the cells, gently re-suspend pellet in another 100 µL of freshly prepared pre-warmed 1×BD Pharmingen Mouse Foxp3 Permeabilization buffer. Incubate for 30 minutes at 37°C in the dark.

9.6.15 Centrifuge 500 x g for 5 minutes and remove buffer.

9.6.16 To wash cells, add 200 µL of stain buffer to each tube, centrifuge 500 g for 5 minutes. Remove buffer.

9.6.17 Add 2 µL of conjugated Foxp3 antibody diluted with stain buffer to re-suspend the pellet. Gently shake or vortex briefly.

9.6.18 Incubate for 20 minutes at RT in the dark.

9.6.19 To wash cells, add 200 µL of stain buffer FBS to each tube, centrifuge 500 g for 5 minutes. Remove buffer.

9.6.20 Re-suspend in 100 µL fixative buffer. Incubate the cells with fixation buffer for 15 to 30 minutes at 2-8°C.

9.6.21 Fixed cells should be washed and suspended in 200 µL Stain buffer. Store the fixed cells at 2-8°C. Next day cells will be acquired by Flow cytometer and analyzed using CytExpert software.

9.6.22 FMO (fluorescence minus one) control will be fixed stained with all the antibodies but one. FMO control will be placed for all different antibodies used in the experiment

9.6.23 Note: Preparation procedure of reagents, is described for fixed volumes and concentrations. However, volume of each reagent will be prepared as per experimental requirement

Panel -2

9.6.24 Prepare cells for flow cytometry staining using sodium azide-free buffers.

9.6.25 Wash cells one time in Dulbecco's Phosphate Buffered Saline (1X DPBS).

9.6.26 Resuspend cells at $1x10^6$ cells/ml in 1X DPBS.

9.6.27 Add 1 µL of the BD Horizon Fixable Viability Stain 510 Stock Solution for each 1million cell suspension (1:1000) and vortex immediately. If required, titrate the dye for optimal performance.

9.6.28 Incubate the mixture for 15 minutes at room temperature protected from light.

9.6.29 Wash cells twice with stain buffer.

9.6.30 Decant the supernatant and gently mix to disrupt the cell pellet.

9.6.31 Stain, fix and permeabilize cells as required.

9.6.32 Discard supernatant and re-suspend the cells in 50 µL of diluted Fc block antibody to each well and incubate for 15 minutes at room temperature.

9.6.33 Surface stain antibody cocktail will be added 50 µL per well after final dilution on top of Fc blocked wells and incubated for 30 minutes at RT in the dark.

9.6.34 Centrifuge the plate at 450 g at 4 ^{o}C for 5 minutes and discard supernatant.

9.6.35 To wash cells, add 200 µL of Stain Buffer FBS to each tube, centrifuge 450 g for 5 minutes. Remove buffer.

9.6.36 Re-suspend in 100 µL fixative buffer. Incubate the cells with fixation buffer for 15 to 30 minutes at 2-8°C.

9.6.37 Fixed cells should be washed and suspended in 200 µL Stain buffer. Store the fixed cells at 2-8°C. Next day cells will be acquired by Flow cytometer and analyzed using CytExpert software.

9.6.38 FMO (fluorescence minus one) control will be fixed stained with all the antibodies but one. FMO control will be placed for all different antibodies used in the experiment

Note: Preparation procedure of reagents, is described for fixed volumes and concentrations. However, volume of each reagent will be prepared as per experimental requirement

9.7. Gating Strategy:

- **Panel 1 (Live cells)**
  - CD3+ CD8+
  - CD3+ CD4+ CD25+ FoxP3+ (T reg)
  - Ratio CD8/T reg
  - CD3+ CD4+
  - CD3+ CD4+ CD25+ (activated)
  - CD3+ CD8+ CD25+ (activated)
  - CD3+ CD8+ NK1.1+ (CD8+ NKT cells)
  - CD3+ CD8- NK1.1+ (NKT cells)
  - CD3- NK1.1+ (NK cells)
- **Panel 2 (Live cells)**
  - CD45+ CD3- F4/80– CD11c+ MHCII+ (Dendritic cells)
  - CD45+ CD3- F4/80+ CD11b+ (Macrophages)
  - CD45- PD-L1+ (Tumour cells)
  - CD45+ CD3+ $CD73^{MFI}$
  - CD45+ CD3- $CD73^{MFI}$
  - CD45- $CD73^{MFI}$

9.8. Data analysis

After gating, Mean ± SD/SEM will be calculated for % total, % parent and number of events. Data generated by CytExpert will be calculated for each group using Microsoft Excel. Raw data obtained from CytExpert will be entered into graph pad prism and significance will be calculated compared to vehicle control as mentioned in Section 11.2

10. BIO-ANALYSIS

Quantitative bio-analysis of [redacted] in the blood samples and tumour will be done using LC-MS/MS. Bioanalysis method optimized for experiment APSL-2022-PH-0005 will be used for this experiment and tentative procedure is mentioned below.

Tumour and Blood processing procedure for PK - [redacted]

Extraction Technique: Protein Precipitation

Extraction Procedure: About 19 µL of blank plasma or tumour homogenate will be transferred into prelabeled micro centrifuge tubes and 1 µL of spiking solution will be spiked to prepare the calibration standards in the range of 0.05-2000 ng/mL .20 µL study samples, calibration standards and quality control samples were crashed with 300 µL Acetonitrile containing internal standard

(carbamazepine/telmisartan/verapamil). The samples will be vortexed for about 2 min. The samples will be centrifuged at 13000 RPM for 5min at 4°C. 200 µL of supernatant will be transferred into HPLC vials and 10 µL will be injected into LC-MS/MS system.

Chromatographic and MS/MS Conditions:

HPLC:	Waters Acquity UPLC
MS:	Triple Quad 6500+
Quantification:	MRM Positive Ion
Q1/Q3 (m/z da.):	358.196/288.1
CC range (ng/mL):	0.05-2000.0

Extraction Procedure	**Protein precipitation with Acetonitrile**		
Column	Agilent ZORBAX SB C18; 4.6*50mm; 5µm		
Mobile Phase	A: 0.1% Formic acid in water		
	B: 0.1% Formic acid in Acetonitrile		
Isocratic Program	**Time (min)**	**A %**	**B %**
	2.00	30	70
Flow	1000 µL/min		
Injection Volume	10 µL		
Run time	2 min		

Compound Parameters:

Compound Code	[redacted]
Q1	358.196
Q3	288.10
Dwell Time (msec)	80.00
DP	166.00
EP	10.00
CE	43.00
CXP	14.00

Source Parameters:

Capillary	7
Cone voltage	30
Source temperature	150
Desolvation temperature	300

11. DATA CAPTURE AND ANALYSIS

11.1. Data capture

In life measurements of body weights, tumour weights and spleen weights will be captured using weighing balance and tumour volumes will be measured using calibrated digital Vernier calipers.

11.2. Data analysis

Body weight, tumour weight & tumour volumes, flow cytometry, and whole blood assay (TNF alpha estimation) data will be calculated as Mean ± SEM or Mean ± SD for each group using Microsoft Excel.

Group comparison statistical analysis will be carried out using GraphPad Prism, version 8.1.1 (or the latest version) or SAS 9.4 (or the latest version). When applicable, results will be presented as Mean ± SEM or as per requirement. Graphs and statistical analysis will be performed using GraphPad Prism or SAS 9.4. For all analyses, failure of assumptions (such as Normality and homogeneity of variance across treatment groups) will be assessed by visual inspection of residual diagnostic plots. Where appropriate, a suitable transformation will be applied to help ensure that the assumptions are valid. For tumour volume, following a log transformation if appropriate, the statistical analysis will be performed either of the change from pre-dose or using pre-dose tumour volume as a covariate, using a two-way analysis of variance (ANOVA) including factors [redacted] (0 or 25 mg/kg/5mL) and aPD1 (0 or 10 mg/kg/10mL) and the interaction between them. The analysis of different days will be performed separately. Comparisons of interest include the effect of [redacted] in the absence and presence of aPD-1, the effect of aPD-1 as a monotherapy and the significance of the interaction. For the analysis of tumour volume, these comparisons will be conducted using a 1–sided 5% test for a reduction in tumour size and will be presented with and without adjustment for multiple comparisons using Dunnett's method. A statistically significant interaction (with the combination showing a larger reduction) is evidence of a synergistic effect in the definition of Bliss. Other parameters tumour weight, body weight and gain, flow cytometry data, and whole blood assay results, will be analyzed using a similar approach and comparisons will be evaluated at the 5% ($p<0.05$) level. Statistically significant differences will be designated by the superscripts throughout the report as stated below:

* - $p< 0.05$, ** - $p<0.01$, *** - $p<0.001$, **** - $p<0.0001$,

Note: SAS 9.4 data analysis will be generated by [redacted] and same will be incorporated into report.

11.3. References

- Bliss (1939), Ann. Appl. Biol. 26:585-615
- Wei et al (2014), J. of Biomolecular Screening. 29:187-821

12. AMENDMENTS AND DEVIATIONS PROCEDURES

The study protocol may be amended or be subjected to alterations. In each case any amendment to the approved study protocol and the reasons for such amendments along with potential impact towards the overall objective and conclusions of the study will be put in writing and realized only after written consent from the study director and sponsor. Any unintended change(s) occurring during the execution of the study will be reported as study protocol deviation report. All amendments and deviations to the study protocol will be documented and reported.

13. REPORT

On completion of the experimental work an excel-based study report containing all raw data will be generated and shared in the shared drive between Aurigene and Heptares Therapeutics Ltd. Independent statistical analysis will be performed or verified by Inferstats Consulting Ltd. (Alderley Park Biohub, Alderley Edge, CHESHIRE SK10 4TG).

14. RECORD RETENTION

The non-electronic raw data derived from this study, study protocol, amendments, deviations if any, raw data and the final report will be stored at Aurigene Pharmaceutical Services, Hyderabad, India up to one year from the date of report finalization/ study completion. The record retention period can be further extended based on mutual consent on a chargeable basis.

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1 LIST OF ABBREVIATIONS

Abbreviation	Full Form / Description
[redacted]	[redacted]
APSL	Aurigene Pharmaceutical Services Limited
CPCSEA	Committee for the Purpose of Control and Supervision of Experiments on Animals
DPBS/ PBS	Dulbecco's phosphate-buffered saline
ELISA	Enzyme Linked Immunosorbent Assay
g	Gram(s)
µg	microgram
GLP	Good Laboratory Practices
IAEC	Institutional Animal Ethics Committee
K_2EDTA	Ethylenediaminetetraacetic acid Potassium salt
mg	Milligram
mL	Milliliter
OVA	Albumin from Chicken Egg
i.p.	Intraperitoneal
IV	Intravenous
po	Per os
[redacted]	Mucosal mast cell protease-1
bid	bis in die
mRNA	messenger Ribonucleic Acid
RT	Room temperature
IgE	Immunoglobulin E
SD	Standard deviation
SOP	Standard operating procedure
TB	Toluidine blue
µL	Microliter

2 STUDY & STUDY PERSONNEL DETAILS

Study No.	:	[redacted]
Study Title	:	Evaluation of [redacted] effect in Food Allergy Model in Female BALB/c mice
Study Director	:	Kishor Chitalkar, M.S. Pharm Department of Pharmacology, Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad-500 049, Telangana, India Tel No.: +91-40-44657340 E-mail: kishordasarath_c@aurigeneservices.com
Head Function	:	D. Ravi Krishna Babu, M. Pharm Department of Pharmacology, Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad-500 049, Telangana, India Tel No.: +91-40-44657345 E-mail: ravikrishna_b@aurigeneservices.com
Responsible Scientists	:	Mr. Ch Sandeep Reddy, M. Pharm Mr. Prashant Yallappa Vadnal, M. Pharm Ms. Sivaranjani J., M. Sc. Mrs. Gouthami Kotha, M. Pharm Mr. Randeep Patro, M. Pharm. Mr. Chatragadda Kishore Kumar
Animal Welfare Officer	:	Dr. Bhavesh Ambalkar M.V.Sc
Test Facility	:	Aurigene Pharmaceutical Services Limited Bollaram Road, Miyapur, Hyderabad-500 049, Telangana, India Telephone No.: + 91-40-44657777 Fax No.: +91-40-44657438
Sponsor	:	[redacted] [redacted]
Total No. of Pages	:	28

3 OBJECTIVE

The objective of this study is to evaluate [redacted] effect in ova induced food allergy model in female BALB/c mice

4 INTRODUCTION

Food allergies are a steadily increasing health and economic problem. Immunologically, food allergic reactions are caused by pathological, allergen-specific Th2 responses resulting in IgE-mediated mast cell degranulation and associated inflammatory reactions. Uptake or penetration of allergen molecules via the respiratory or gastrointestinal epithelia results in uptake, processing and presentation of allergen-derived peptides by antigen presenting cells (APCs). Activated APCs express co-stimulatory signals and secrete cytokines that promote the differentiation of naïve antigen-specific T cells into effector cells. Naive, allergen-specific T cells activated by these APCs in turn differentiate into T helper 2 (Th2) cells, directed by cytokines (e.g., interleukin 25 (IL-25) or thymic stromal lymphopoietin (TSLP)) or damage associated molecular patterns (DAMPs) released from damaged epithelial cells (e.g., IL-33), which subsequently trigger the local activation of innate like lymphocytes type II cells (ILC2s). Upon activation, these ILC2s provide cytokines IL-13 and IL-4 driving Th2 activation and differentiation.

Once activated, allergen-specific Th2 cells enable activation, differentiation, and isotype switching of allergen-specific B cells towards IgE-producing plasma cells. The produced allergen-specific IgE antibodies subsequently bind to the high affinity IgE receptor [redacted] on the surface of mast cells and basophils, resulting in their sensitization towards the allergen

Upon second contact with the allergen, cross-linking of surface-bound IgE antibodies results in mast cell and eosinophil activation and local release of histamine, cytokines, proteases, hormones, and other pro-inflammatory mediators in a process called degranulation which then causes the allergic inflammation and the associated symptoms

Therefore, the allergic type I reaction is characterized by a Th2-mediated IgE production resulting in mast cell activation and expansion. In affected food allergic patients, these processes lead to a local inflammation of the mouth mucosa, the face, the throat, and are

often paralleled by skin reactions. Furthermore, food allergic patients suffer from abdominal pain, nausea, diarrhea, or vomiting caused by local inflammation of the gastrointestinal tract. In addition, if initiated, [redacted] can result in acute life-threatening anaphylactic reactions

Clinically, food allergies are characterized by local inflammation of the mouth mucosa, the face, the throat, the gastrointestinal tract, are frequently paralleled by skin reactions, and can result in life-threatening anaphylactic reactions. To better understand food allergies and establish novel treatment options, mouse models are indispensable. In the present study, [redacted] will be evaluated in ova induced food allergy model in female BALB/c mice.

5 REGULATORY COMPLIANCE AND APPROVALS

5.1 Test Guidelines

This study is not based on any specific test guidelines. However, it will be conducted in compliance with the mutually agreed study plan & SOPs of Aurigene Pharmaceutical Services Limited.

5.2 Good Laboratory Practice (GLP) Compliance

The study will be conducted as a non-GLP study and the procedures will be carried out as per the current standard operating procedures of the test facility and Study Plan.

5.3 Institutional Animal Ethics Committee (IAEC) Approval

This study will be performed in conformity with the Animal Welfare Law under the Animal License No. 3/PO/RcBi-S/RcBi-L/1999/CPCSEA and as per the approval of IAEC protocol No: IAEC/AUR/2023/08/50/PH/Ova food allergy/PV/145/M/F-96.

6 SAFETY PRECAUTIONS

Gloves, cap, face mask, gowns or lab coats and/or other appropriate personnel protective equipment will be used to handle the test and /or standard compound to ensure adequate personal health and safety and to avoid inhalation and skin contact with the test item. In case of eye contact, the eyes must be washed thoroughly with water and medical treatment to be sought. In case of skin contact, skin must be washed with soap and water with subsequent medical aid. Leak and spillage procedures: area of spillage will be ventilated; appropriate personnel protective equipment (gloves, cap, face mask, gowns and safety glasses) will be worn before entering the area. Area including floor and walls are flushed with water, wet mop or vacuum pump will be

used to pick up spilled product. Unnecessary and unprotected personnel are kept away from entering in the spilled area.

7 TENTATIVE STUDY SCHEDULE

Acclimatization	9th Aug 2023
Randomization	14th Aug 2023
OVA sensitization	16th and 30th Aug 2023
[redacted] and [redacted] dosing	12th Sept 2023
OVA Challenge	13th Sept 2023
In life completion date	21st Sept 2023
Serum level estimation of IgE, anti OVA IgE, mMCP1	26th Sept 2023
Expression of gene analysis	30th Sept 2023
Tissue mast cell count	To be updated

8 MATERIALS AND METHODS

8.1 Test Item

Compound: [redacted] ([redacted]

Batch no: [redacted]

Formula weight: [redacted]

Molecular weight: [redacted]

Salt type: NA

Purity: 99.28%

Salt correction: NA

Purity correction: 1.007

Final correction factor: 1.007

8.2 Materials and Reagents

Name	Make	Catalog No.	Batch/ Lot No
DPBS	Gibco	14190-144	2436266
Imject Alum	Thermofisher Scientific	77161	WF322465

OVA (Albumin from chicken egg white)	Sigma	A5503-25G	SLCK7421
[redacted]	inVivoMab – BioXcell	BE0293	851122J2
RNeasy mini kit	Qiagen	74106	172033731
High capacity cDNA Reverse Transcription Kit	Applied Biosystems	4368814	2656643
TaqMan™ Fast Advanced Master Mix	Applied Biosystems	4444557	01253958
Total Ig E detection kit	Chondrex	3005	TBU
Mouse Anti OVA IgE antibody Assay kit	Chondrex	3004	TBU
[redacted] ([redacted] Mouse Uncoated ELISA Kit	Invitrogen	[redacted]	[redacted]
TaqMan Assay, [redacted] gene, Assay ID: [redacted]	Thermofisher Scientific	[redacted]	[redacted]
TaqMan Assay, [redacted] gene, Assay ID: [redacted]	Thermofisher Scientific	[redacted]	[redacted]
TaqMan Assay, [redacted] gene, Assay ID: [redacted]	Thermofisher Scientific	4448489	TBU
GAPDH gene, Assay ID: m99999915_g1	Thermofisher Scientific	4331182	168554
Recombinant Anti-[redacted] antibody	Abcam	[redacted]	[redacted]
Rabbit specific IHC polymer detection kit HRP/DAB	Abcam	ab209101	GR3433022-1
DAB substrate kit	Abcam	ab64238	GR3413290-1
Blocking reagent	Chemcruz	sc-516214	A2123
Limonene mount	Chemcruz	sc-45087	L0617
Citrate Concentrated Solution	Chemcruz	sc-294091	B1521

Note: The experiment is planned to be conducted using the above mentioned make/ catalog no./Batch/ Lot No and any changes will be captured in lab note book

8.3 Instruments

Name	Make
Homogenizer	Omni Bead Ruptor elite
NanoDrop One	Thermo Scientific
Proflex PCR system	Applied Biosystem
ABI 7500 Thermal Cycler	Applied Biosystems
Micro plate ELISA reader	Bioteck Powerwave XS
Rodent thermometer BIO-TK9882	Bioseb

8.4 Test System and Environment

Species and strain	BALB/c mice
Number of animals in study	48
Initial Age and approximate body weight	6-7 Weeks
Sex	Female
Breeder/ Supplier	Vivo Biotech Limited
Animal housing	Animals will be housed in group of 4 in polypropylene cages containing irradiated corncob bedding material, with enrichment toys. Room temperature and relative humidity ranges of 22 ± 2° C and 50 ± 20%, respectively. Twelve-hour light/dark cycle provided via automatic timer. Ventilation will produce 10-15 air changes/ hour.
Feed and water	Animals will be fed with the Gamma Irradiated commercial rodent maintenance diet 1324. Purified water will be provided *ad libitum* in polypropylene bottles.
Cage and Individual animal Identification	Cage card indicating animal number, sex, study number and different dose groups, tail marking for individual animal identification.

On day 0 and 14, animals in group 1 will be sensitized by intraperitoneal injection of DPBS and for group 2 to 5, 50 μg of egg white albumin (OVA) and 2 mg of aluminum hydroxide and magnesium hydroxide (Alum) in saline will be injected.

After sensitization, from day 27 to day 36 animals in the respective groups will be orally dosed twice every day with vehicle / [redacted] (3 mpk, 10 mpk & 30 mpk). On the day of ova challenge [redacted] will be administered 4 hours before ova administration and after 1-2 hours of fasting initiation. Every alternate day from day 27 to day 35, mice in group 3 will be dosed with 0.1 mg [redacted] First dose will be administered intravenously while the rest of the four doses will be administered intra peritoneally.

Every alternate day from day 28 to day 36 animals in all groups will be orally administered 250 μL of sterile saline containing 50 mg of OVA. Before each intragastric challenge, mice will be deprived of food for 5-6 hours with the aim of limiting antigen degradation in the stomach.

Diarrhea will be assessed by visually monitoring mice for up to ~60 min following intragastric challenge. Mice demonstrating loose or liquid stools will be recorded as diarrhea-positive animals. Rectal temperature will be measured with a digital thermometer at 0, 10, 20, 30 and 60 min after intragastric OVA challenge.

On day 36 or termination day, ~4-5-hour post ova dosing, blood sample will be collected under Isoflurane anesthesia. Plasma will be collected to estimate compound levels. Serum will be collected to estimate the levels of IgE, anti OVA IgE and mMCP1 by ELISA. After blood collection animals will be sacrificed, stomach and intestine will be collected and stored in 10% NBF/ and processed by standard histological techniques for mast cell count along with IHC.

Note: Based on the disease induction assessed by Diarrhea, study shall be terminated on Day 36. However, based on animal response study may be terminated earlier or extended.

9.2.1 Preparation of vehicle and OVA formulation:

9.2.2.1 DPBS/ vehicle preparation

Ready to use DPBS will be used.

9.2.2.2 OVA WITH Alum intraperitoneal formulation

2.5 mg of OVA will be dissolved in 7.5 ml of DPBS and mixed well. To this ova solution 2.5mL imject alum will be added.

Shake the capped bottle of Imject Alum well before use. Add 2.5 ml of Imject Alum dropwise with constant mixing to the ova solution so the final volume ratio of Imject Alum to immunogen is 1:3. Continue mixing for 30 minutes after adding the Imject Alum. Mixing allows the Imject Alum to effectively adsorb antigen. Each animal in Group 2 to 6 will receive 200µL of formulation during sensitization period

Note: Imject Alum, 50mL, contains an aqueous solution of aluminum hydroxide (40mg/mL) and magnesium hydroxide (40mg/mL) plus inactive stabilizers

9.2.2.3 OVA oral formulation

2.8 g of OVA will be weighed and mixed in 9.00 ml of DPBS. After thorough mixing volume will be adjusting to 14 mL using DPBS. Each animal in all groups will receive 250µl of formulation during evaluation period.

9.2.2.4 Preparation of 1% w/v of Methylcellulose:

1 gram of Methylcellulose will be weighed, transferred into a glass container and dissolved in purified water (50 mL) by stirring over magnetic stirrer. The contents will be transferred, and final volume will be made up to 100 mL with water in a suitable measuring cylinder to get strength of 1% w/v methylcellulose stock solution. The prepared solution will be stored in a suitable glass bottle.

9.2.2.5 Preparation of 1% w/v Tween80 in Water:

1 gram of Tween 80 will be weighed in a glass container and dissolved in purified water (50 mL) by stirring over magnetic stirrer. The contents will be transferred, and final volume will be made up 100 mL with water in a suitable measuring cylinder to get strength of 1% Tween 80 stock solution. The prepared solution will be stored in a suitable glass bottle.

9.2.2.6 Preparation of vehicle for Oral dose:

Vehicle composition: 0.5% Tween 80 + 0.5% Methylcellulose

6 mL of vehicle will be prepared by mixing 3 mL of 1% Tween 80 + 3 mL of 1% Methylcellulose

9.2.2.7 [redacted] 3 mg/kg P.O dose preparation

1.20 mg of [redacted] (correction factor = 1.007) will be weighed and transferred into a clean mortar. 2 ml of 1% Tween 80 will be added dropwise and the compound will be triturated using a pestle to form a fine suspension. Then 2 mL of 1% of methylcellulose will be added.

9.2.2.8 [redacted] 10 mg/kg P.O dose preparation

~3.02 mg of [redacted] (correction factor = 1.007) will be weighed and transferred into a clean mortar. 1.5 ml of 1% Tween 80 will be added dropwise and the compound will be triturated using a pestle to form a fine suspension. Then 1.5 mL of 1% of methylcellulose will be added.

9.2.2.9 [redacted] 30 mg/kg P.O dose preparation

~9.06 mg of [redacted] (correction factor = 1.007) will be weighed and transferred into a clean mortar. 1.5 ml of 1% Tween 80 will be added dropwise and the compound will be triturated using a pestle to form a fine suspension. Then 1.5 mL of 1% of methylcellulose will be added.

9.2.2.10 0.1 mg/[redacted]

~109µL [redacted] stock solution (9.13mg/mL) will be diluted to 2 mL with DPBS to get a concentration of 0.5 mg /mL solution. 200 µL of 0.1 mg /mL [redacted] solution will be administered to each animal in group 3.

Note: Preparation procedure of solutions, vehicle and other formulations is described for fixed volumes. However, volume of each solution or vehicle or formulation will be prepared as per experimental requirement. [redacted] preparation shall change based on concentration of stock.

9.3 Measurements and Observations

9.3.1 Animal Observation for Mortality, Clinical Signs of Toxicity

Animals will be observed for mortality, morbidity and for any obvious clinical signs during experiment.

9.3.2 Body Weight

Animal body weight will be measured and recorded twice in a week during immunization period. After [redacted] and [redacted] administration initiation, everyday body weight will be measured and recorded.

9.3.3 Body temperature

Rectal temperature will be measured with a digital thermometer every 10 minutes up to 30 min and at 60 min after intragastric OVA challenge.

9.3.4 Diarrhea

Diarrhea will be assessed by visually monitoring mice for up to 60 min following intragastric challenge. Mice demonstrating loose or liquid stools will be recorded as diarrhea-positive animals.

9.3.5 Blood collection for analysis

9.3.5.1 Plasma collection: On day 36(Day of study termination), ~5-hour post dosing animals will be anaesthetized with isoflurane and blood sample will be collected into K_2EDTA containing 1.5 mL micro centrifuge tubes. Blood samples will be centrifuged at 6000 rpm, 4°C for 5 min, plasma separated and stored at -60° C to -80°C until analysis of compound levels.

~500 µL of blood will be collected to separate the serum for estimating of levels of IgE, anti OVA IgE and mMCP1 by ELISA in 1.5 mL micro centrifuge tube from all animals. After collection, tubes will be kept in slanting position for 30 to 60 mins, and tubes will be centrifuged at 6000 rpm for 10 min at 4°C for serum collection. Samples will be stored at -60° to -80°C until analysis

9.4 Study Termination and tissue collection

After blood collection, animals will be sacrificed for tissue collection. Ear, Stomach and intestine will be collected, stored in 10% NBF and/or Carnoy's solution and processed by standard histological techniques for mast cell count. Separately some part of ear, stomach and intestine will be collected in tri reagent for expression of [redacted] [redacted] and [redacted] gene analysis.

9.5 Quantification of Serum Anti-OVA IgE levels by ELISA

9.5.1 Preparation of OVA Solution:

Dilute one vial of Ovalbumin (OVA) with 10 ml of OVA Dilution Buffer. Alternatively, dilute according to the table below. Any leftover OVA Stock Solution may be stored at -20°C for future assays.

Strip #	OVA (µl)	Solution A (ml)
2	17	1.7
4	33	3.3
6	50	5.0
8	66	6.6
10	82	8.2
12	100	10.0

9.5.2 Preparation of Standard Dilutions:

9.5.2.1 The recommended standard range is 0.8-50 ng/ml.

9.5.2.2 Dissolve one vial of Standard in 1 ml of Sample/Standard Dilution Buffer (Solution B) and keep it as a 1000 ng/ml standard stock.

9.5.2.3 Take 50 µl of the standard solution (1000 ng/ml) and add it to 950 µl of Solution B to make a 50 ng/ml IgE standard solution.

9.5.2.4 Then serially dilute it with Solution B. For example, mix 250 µl of the stock solution with an equal volume of Solution B to make a 25 ng/ml stock solution.

9.5.2.5 Repeat it five more times for 12.5, 6.3, 3.1, 1.6, 0.8 ng/ml standard solutions.

9.5.2.6 The remaining 1000 ng/ml standard stock solution can be stored at -20^{0}C for use in a future assay.

9.5.3 Sample Dilutions:

If required samples will be diluted 1:10 or more in Sample/Standard Dilution Buffer (Solution B)

9.5.4 Preparation of Secondary Antibody:

Dissolve one vial of Secondary Antibody in 10 ml Secondary Antibody Dilution Buffer (Solution C). Alternatively, dissolve one vial of Secondary Antibody in 50 µl of Solution C and dilute according to the following table.

Strip #	Secondary Antibody (µl)	Solution C (ml)
2	8	1.7
4	17	3.3
6	25	5.0
8	33	6.6
10	42	8.2
12	50	10.0

9.5.5 Preparation of Streptavidin Peroxidase:

Dilute one vial of Streptavidin Peroxidase in 10 ml Streptavidin Peroxidase Dilution Buffer (Solution D).

Strip #	Streptavidin Peroxidase (µl)	Solution D (ml)
2	8	1.7
4	17	3.3
6	25	5.0
8	33	6.6
10	42	8.2
12	50	10.0

9.5.6 Preparation of TMB Solution:

Use new tubes when preparing TMB solution. Just prior to use, prepare TMB solution with Chromogen Dilution Buffer as shown in the following table

Strip #	TMB (µl)	Chromogen Dilution Buffer (ml)
2	34	1.7
4	66	3.3
6	100	5.0
8	132	6.6
10	164	8.2
12	200	10.0

9.5.7 Preparation of Wash buffer: Dilute 50 ml of 20X wash buffer in 950 ml of distilled water (1X wash buffer).

9.5.8 Assay Procedure:

9.5.8.1 Add 100 µl of diluted OVA into wells Incubate at 4 degrees Celsius overnight.

9.5.8.2 Wash the plate with 1X wash buffer at least 3 times. Empty the plate by inverting it and blotting on a paper towel to remove excess liquid. Do not allow the plate to dry out.

9.5.8.3 Add 100 µl of diluted standards and samples into wells Incubate at room temperature for 2 hours.

9.5.8.4 Wash the plate as per step 9.5.8.2.

9.5.8.5 Add 100 µl of diluted secondary antibody solution into wells Incubate at room temperature for 1 hour.

9.5.8.6 Wash the plate as per step 9.5.8.2.

9.5.8.7 Add 100 µl of diluted streptavidin peroxidase solution into wells Incubate at room temperature for 1 hour.

9.5.8.8 Wash the plate as per step 9.5.8.2.

9.5.8.9 Add 100 µl of TMB solution into wells Incubate at room temperature for 5-25 minutes.

9.5.8.10 Add 50 µl of 2N sulphuric acid (stop solution) into each well

9.5.8.11 After zeroing the plate reader against the blank, read optical density at 450nm. If the plate reader is not capable of adjusting for the blank, manually subtract the mean OD of the substrate blank from all readings

9.6 Quantification of Serum Total IgE levels by ELISA

9.6.1 Preparation of Capture Antibody:

Dilute one vial of Capture Antibody with 10 ml of Capture Antibody Dilution Buffer (Solution A). Alternatively, dilute according to the table below. Any leftover Capture Antibody Stock Solution may be stored at -20°C for future assays.

Strip #	Capture Antibody (µl)	Solution A (ml)
2	17	1.7
4	33	3.3
6	50	5.0
8	66	6.6
10	82	8.2
12	100	10.0

9.6.2 Preparation of Standard Dilutions

9.6.2.1 The recommended standard range is 1.6-100 ng/ml.

9.6.2.2 Dissolve one vial of Standard (100 ng/vial) in 1 ml of Sample/Standard Dilution Buffer (Solution B) to make a 100 ng/ml IgE standard stock solution.

9.6.2.3 Then, serially dilute it with Solution B. For example, mix 250 µl of the standard (100 ng/ml) with an equal volume of Solution B to make a 50 ng/ml solution.

9.6.2.4 Repeat it five more times to prepare 25, 12.5, 6.3, 3.1, and 1.6 ng/ml standard solutions.

9.6.2.5 The remaining 100 ng/ml standard stock can be stored at -20°C for use in a second assay

9.6.3 Preparation of Sample Dilutions

If required samples will be diluted 1:10 or more in Sample/Standard Dilution Buffer (Solution B)

9.6.4 Preparation of Wash Buffer

Dilute 50 ml of 20X wash buffer in 950 ml of distilled water (1X wash buffer).

9.6.5 Preparation of Detection Antibody

Dissolve one vial of Detection Antibody in 10 ml Detection Antibody Dilution Buffer (Solution C). Alternatively, dissolve one vial of Detection Antibody in 50 µl of Solution C and follow the dilutions in the following table.

Strip #	Detection Antibody (µl)	Solution C (ml)
2	8	1.7
4	17	3.3
6	25	5.0
8	33	6.6
10	42	8.2
12	50	10.0

9.6.6 Preparation of Streptavidin Peroxidase

Dilute one vial of Streptavidin Peroxidase in 10 ml of Streptavidin Dilution Buffer (Solution D).

Strip #	Streptavidin Peroxidase (µl)	Solution D (ml)
2	8	1.7
4	17	3.3
6	25	5.0
8	33	6.6
10	42	8.2
12	50	10.0

9.6.7 Preparation of TMB

Use new tubes when preparing TMB. Dilute one vial of TMB with 10 ml of Chromogen Dilution Buffer just prior to use.

Strip #	TMB (µl)	Chromogen Dilution Buffer (ml)
2	34	1.7
4	66	3.3
6	100	5.0
8	132	6.6
10	164	8.2
12	200	10.0

9.6.8 Assay Procedure:

9.6.8.1 Add 100 µl of diluted capture antibody into wells Incubate at 4 degrees Celsius overnight.

9.6.8.2 Wash the plate with 1X wash buffer at least 3 times. Empty the plate by inverting it and blotting on a paper towel to remove excess liquid. Do not allow the plate to dry out.

9.6.8.3 Add 100 µl of diluted standards and samples into wells Incubate at room temperature for 2 hours.

9.6.8.4 Wash the plate as per step 9.6.8.2.

9.6.8.5 Add 100 µl of diluted detection antibody solution into wells Incubate at room temperature for 1 hour.

9.6.8.6 Wash the plate as per step 9.6.8.2.

9.6.8.7 Add 100 µl of diluted streptavidin peroxidase solution into wells Incubate at room temperature for 1 hour.

9.6.8.8 Wash the plate as per step 9.6.8.2.

9.6.8.9 Add 100 µl of TMB solution into wells Incubate at room temperature for 5-25 minutes.

9.6.8.10 Add 50 µl of 2N sulphuric acid (stop solution) into each well

9.6.8.11 After zeroing the plate reader against the blank, read optical density at 450nm. If the plate reader is not capable of adjusting for the blank, manually subtract the mean OD of the substrate blank from all readings

9.7 Quantification of Serum [redacted] levels by ELISA

9.7.1 Preparation of Coating Buffer (1X)

Make a 1:10 dilution of PBS (10X) in deionized water.

9.7.2 Preparation of Capture Antibody

Dilute capture antibody (250X) 1:250 in Coating Buffer (1X).

9.7.3 Preparation of 5X ELISA/ELISPOT Diluent

Dilute Diluent Concentrate (5X) 1:5 in deionized water.

9.7.4 Preparation of Standard

Reconstitute mouse [redacted] standard by addition of distilled water. Reconstitution volume is stated on the label of the standard vial. Allow the standard to reconstitute for 10–30 minutes. Swirl or mix gently to ensure complete and homogeneous solubilization (concentration of reconstituted standard = 15,000 pg/mL). Mix well prior to making dilutions. The standard has to be used immediately after reconstitution and cannot be stored.

9.7.5 Preparation of Detection Antibody

Dilute detection antibody (250X) 1:250 in ELISA/ELISPOT Diluent (1X).

9.7.6 Preparation of Enzyme

Dilute HRP Concentrate (250X) 1:250 in ELISA/ELISPOT Diluent (1X).

9.7.7 Experimental procedure

Note: Shaking is recommended for all incubation steps.

Note: In case of incubation without shaking, the obtained O.D. values may be decreased.

9.7.7.1 Add 100 µl of diluted capture antibody into wells Incubate at 4 °C overnight.

9.7.7.2 Aspirate wells and wash 3 times with >250 µL/well Wash Buffer. Allowing time for soaking (~1 minute) during each wash step increases the effectiveness of the washes. Empty the plate by inverting it and blotting on a paper towel to remove excess liquid. Do not allow the plate to dry out.

9.7.7.3 Add 200 µl of ELISA/ELISPOT Diluent (1X) Incubate at room temperature for 1 hour

9.7.7.4 Aspirate and wash at least once with Wash Buffer as per step 9.7.7.2.

9.7.7.5 Perform 2-fold serial dilutions of the top standards to make the standard curve for a total of 8 points. For that add 100 µL of ELISA/ELISPOT Diluent (1X) to the wells leaving the ęrst wells empty. Add 200 µL/well of top standard concentration to the ęrst empty

wells A1/A2. Transfer 100 µL of top standard from wells A1/A2 to wells B1/B2. Mix the contents of the wells B1 and B2 by repeated aspiration and ejection and transfer 100 µL to wells C1/C2. Take care not to scratch surface of the microwells. Continue this procedure 5 times. Add 100 µL of ELISA/ELISPOT Diluent (1X) for blank

9.7.7.6 Add 100 µl of samples into appropriate wells.

9.7.7.7 Seal the plate and incubate at room temperature for 2 hours (or overnight at 4°C for maximum sensitivity)

9.7.7.8 Aspirate wells and wash 3 times as per step 9.7.7.2.

9.7.7.9 Add 100 µl of diluted detection antibody solution into wells Incubate at room temperature for 1 hour.

9.7.7.10 Aspirate wells and wash 3 times as per step 9.7.7.2.

9.7.7.11 Add 100 µL/well of diluted Avidin-HRP. Seal the plate and Incubate at room temperature for 30 minutes.

9.7.7.12 Aspirate wells and wash 3 times as per step 9.7.7.2.

9.7.7.13 Add 100 µl of TMB solution into wells Incubate at room temperature for 5-15 minutes.

9.7.7.14 Add 100 µl of stop solution into each well

9.7.7.15 Read plate at 450 nm. If wavelength substraction is available, substract the values of 570 nm from those of 450 nm and analyze data.

9.8 Gene expression analysis:

Gene expression analysis of [redacted] and GAPDH in tissue will be carried out as per the below mentioned procedure

9.8.1 RNA Isolation:

9.8.1.1 RNA will be isolated from the tissues by using RNeasy Mini Kit (Qiagen) according to manufacturer's instructions with brief details mentioned as below.

9.8.1.2 Tissues (~100mg) will be taken out from -80^{0}C and keep on ice and 1 ml of cold TRI reagent will be added. Tissue will be homogenized completely under ice cold conditions.

9.8.1.3 Blood samples added to TRI reagent will processed after 5 min incubation at room temperature as mentioned in bellow procedure for tissue homogenates.

9.8.1.4 200ul of chloroform will be added to homogenates, and samples will be vortex vigorously for 10-15 seconds and incubate at RT for 10 minutes.

9.8.1.5 After incubation, tube will be centrifuged at 12000xg / 15 min / 4°C.

9.8.1.6 Aqueous phase (RNA) upper layer of 200µl will be collected into a new pre-chilled tube, without disturbing the interphase (DNA).

9.8.1.7 1 ml of 70% Ethanol (ice cold preferably) will be added to the aqueous layer and mix gently by inverting tubes for 10 seconds.

9.8.1.8 Sample will be transferred (700ul at a time) to RNeasy mini spin column (binding capacity of 100µG).

9.8.1.9 After transfer, tube will be centrifuged at 8000xg / 60 seconds / 4°C, and discard flow through.

9.8.1.10 700ul of RW1 buffer will be added and centrifuge 8000xg / 60 seconds / 4°C, discard flow through.

9.8.1.11 500ul of RPE buffer will be added and centrifuge at 8000xg / 60 seconds / 4°C, and discard flow through.

9.8.1.12 Repeat above step

9.8.1.13 Dry spin will be given at 8000xg / 60 seconds / 4°C, and discard flow through.

9.8.1.14 Collection tube will be changed and 30-50µl of RNase free water will be added and incubated at RT for 1 to 2 minutes and centrifuge at 8000xg / 60 seconds / 4°C.

9.8.1.15 RNA quantity will be estimated by using Nanodrop.

9.8.2 cDNA Synthesis:

9.8.2.2 Based on RNA concentrations of samples and manufacturer's instructions cDNA will be prepared for 2µg/20µl by using High Capacity cDNA reverse transcription kit (applied Biosystems).

9.8.2.3 2X Reverse Transcription Master Mix will be prepared (per 20 µl reaction) as mentioned below.

Component	Volume/Reaction (µL)
10X RT Buffer	2.0
25X dNTP Mix (100 mM)	0.8
10X RT Random Primers	2.0
MultiScribe™ Reverse Transcriptase	1.0
RNase Inhibitor	1.0

Compilation will be done for every target gene by using MS-Excel and GraphPad Prism software.

9.9 [redacted] staining and counting using Toludine blue/Alcian blue stain

9.9.1 Tissue samples fixed in formalin/ carnoy solution are processed and embedded in paraffin blocks and sections of 4µM thickness will be taken using microtome and mounted on slides. Tissues will be stained using Toluidine blue stain or Alcian blue stain

9.9.2 **Tissue [redacted] counting:** Stained tissues will be counted for mast cell under microscope in 5 different locations in each tissue.

9.10 Tissue [redacted] staining and counting using Immunohistochemistry:

9.10.1 Preparation of Reagents for IHC:

9.10.1.1 **Preparation of Citrate buffer pH 6:**

10mL of 10X citrate buffer will be diluted to 90mL using milli Q water and pH will be adjusted to 6 using 1N HCL/NaoH.

9.10.1.2 **Dilution of [redacted] (100X/1000X):** 5µL of [redacted] antibody will be diluted to 500µL (100X) using 0.1% BSA in PBS.

9.10.1.3 **Preparation of Rinse buffer:** DAB150 kit contains 20X rinse buffer. 7.5ml of 20X rinse buffer will be diluted to 1X rinse buffer by addition of 142.5mL of water.

9.10.1.1 **Preparation of 1% hydrogen peroxide:** To 2mL of 30% hydrogen peroxide (H_2O_2) add 58mLof water to get a concentration of 1% hydrogen peroxide

9.10.1.2 **Chromogen Reagent preparation**

To prepare DAB chromogen reagent add 30µL of DAB chromogen to 1.5mL of DAB substrate

9.10.2 Deparaffinization

NOTE: Do not allow slides to dry at any time during this procedure.

- Incubate sections in two washes of xylene for 5 min each.
- Incubate sections in one wash of 1:1 Xylene alcohol solution for 5 min
- Incubate sections in two washes of 100% alcohol for 3 min each.
- Incubate sections in two washes of 95% alcohol for 3 min each.
- Incubate sections in two washes of 70% alcohol for 3 min each

- Incubate sections in two washes of 50% alcohol for 3 min each
- Wash sections two times in dH_2O for 1 min each

9.10.3 Antigen Unmasking:

- Heat slides in a microwave submersed in 1X citrate unmasking solution until boiling is initiated; follow with 20 min at a sub-boiling temperature (95°-98°C). Cool slides on bench top for 30 min.

9.10.4 Staining:

- Immerse the slides in 1% H2O2 for 10 mins
- Wash the slides by incubating in Rinse buffer for two times 5 min each
- Blot dry the area around the tissue and then use a PAP Pen to create a hydrophobic barrier to contain the Staining Reagents.
- Allow the PAP Pen barrier to dry for at least 2 minutes at room temperature but ensure that the tissue remains hydrated.
- Note: Leave space between the tissue and the PAP Pen barrier as the PAP Pen solution may adversely affect staining of the tissue, causing false negative results.
- Apply the blocking reagent to the sections and incubate in an enclosed chamber for 1hr.
- Wash the slides by incubating in rinse buffer once for 5 min.
- Apply primary antibody or antibody diluent (negative control) over the entire sections and incubate in an enclosed chamber at 2-8°C overnight.
- Wash the slides three times in rinse buffer for 5min each
- Apply sufficient amplifier and incubate in an enclosed chamber for 30 minutes.
- Wash the slides three times in rinse buffer for 5min each
- Apply sufficient detector and incubate in an enclosed chamber for 30 minutes
- Wash the slides three times in rinse buffer for 5min each
- Apply sufficient DAB (chromogen reagent) to cover entire section and incubate in an enclosed chamber for 10 minutes.
- Wash the slides in water.

- Apply the 25% Hematoxylin counterstain solution to the sections and incubate in an enclosed chamber for 30 sec.
- Wash the slides in water.
- Place the tissue slides directly into a container filled with deionized water until mounting.

9.10.5 Dehydration of Sections:

- Incubate sections in 50% alcohol two times for 10 sec each.
- Repeat in 70% alcohol, incubating sections two times for 10 sec each.
- Repeat in 95% alcohol, incubating sections two times for 10 sec each.
- Repeat in 100% alcohol, incubating sections two times for 10 sec each.
- Repeat in xylene, incubating sections two times for 10 sec each.
- Mount sections with coverslips and mounting medium

10 STATISTICAL ANALYSIS

Graphs and statistical analysis will be performed using students-t test, One-way ANOVA and Two-way ANOVA followed by post hoc test with GraphPad Prism 8 (GraphPad Software). Wherever applicable, results will be presented as a Mean ± SD.

11 DATA PRESENTATION AND REPORTS

Upon completion of study, electronic copy of data in the form of power point presentation will be shared along with raw data (MS EXCEL) for sponsor review. If required, based on sponsor request draft report will be prepared and shared with sponsor for review. After finalization of report by sponsor, signed copy of original report will be send to sponsor. All raw data (or authenticated copies thereof) and a copy of the final report, original study plan and amendments will be retained in the archive for a period of 2 years, by then the sponsor needs to indicate further storage at additional cost

12 ANIMAL ETHICS

Animals considered to be in pain, distress or moribund will be examined by authorized personnel (qualified technicians, study director or deputy, experienced veterinarian) and, if deemed

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1. LIST OF ABBREVIATIONS

Abbreviation		Full Form/Description
~	:	Approximately
%	:	Percentage
^{0}C	:	Degree Centigrade
ANOVA	:	Analysis of Variance
CIA	:	Collagen induced Arthritis
CPCSEA	:	Committee for the Purpose of Control and Supervision of Experiments on Animals
FDA	:	Food and Drug Administration
GLP	:	Good laboratory practice
gm	:	Gram
h	:	Hour
H&E	:	Haematoxylin and Eosin
IAEC	:	Institutional Animal Ethics Committee
IFA		Incomplete Freund's adjuvant
IL-1β	:	Interleukin 1 beta
IL-6		Interleukin 6
Mg/kg	:	Milligram/kilogram
mg/mL	:	Milligrams/ millilitre
min	:	Minutes
mL	:	Millilitre
qd	:	Once a day
RA	:	Rheumatoid Arthritis
rpm	:	Revolutions Per Minute
RT	:	Room Temperature
s.c	:	Subcutaneous
SBEBCD		Sulfobutyl-ether-β-cyclodextrin sodium salt
SD	:	Standard Deviation
SOP	:	Standard Operating Procedure
TNF-α	:	Tumor necrosis factor alpha
w/v	:	Weight per unit volume
µL	:	Micro litre

2. STUDY & STUDY PERSONNEL DETAILS

Study No.	:	[redacted]
Study Title	:	Effect of [redacted] & [redacted] in Collagen Induced Arthritis in female SD rats
Study Director	:	Prashant Yallappa Vadnal M. Pharm. Pharmacology Department Aurigene Pharmaceutical Services Limited Bollaram Road, Miyapur Hyderabad-500, 049, Telangana, India Telephone No.: + 91-40-44657340 Email: prashantyallappa_v@aurigeneservices.com
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Sponsor	:	[redacted] Boston, MA 022101 anas.fathallah@lapixtherapeutics.com
Responsible Scientists	:	Mr. Kishor Chitalkar, M. Pharm Mr. Ch Sandeep Reddy, M. Pharm Mr. Sandeep Mahankali M. Pharm Dr. Arabinda Adak M.V.Sc.
Animal Welfare Officer	:	Dr. Sachin Bhalchandra Pagrut M.V. Sc

3. OBJECTIVE

The objective of the study is to evaluate the effect of [redacted] & [redacted] in Collagen Induced Arthritis in Female SD rats

4. INTRODUCTION

Rheumatoid arthritis (RA) is a chronic and genetically complex inflammatory disorder that leads to erosive destruction of peripheral joints with a considerable impact on quality of life. The etiology and pathophysiology of RA are not clearly understood, and many cell types, such as fibroblasts, T cells, B cells, monocytes/macrophages and dendritic cells (DCs), have been implicated. These inflammatory cells infiltrate the synovium and are further activated to release cytokines, autoantibodies, and matrix metalloproteinase (MMP), leading to cartilage and bone destruction1. Pro-inflammatory cytokines produced by Th17 cells (e.g., interleukin (IL)-17) and macrophages (e.g., tumor necrosis factor-α (TNF-α), IL-1β, and IL-6) play significant roles in mediating joint inflammation.

Collagen induced arthritis (CIA) model in rats is a widely used animal model of human arthritis as it shares immunological and pathological similarities to human Rheumatoid Arthritis (RA).

In the present study [redacted] & [redacted] will be tested for its efficacy in Collagen induced arthritis in female SD rats

5. REGULATORY COMPLIANCE AND APPROVALS

5.1 Test Guidelines

This study is not based on any specific test guidelines. However, it will be conducted in compliance with the mutually agreed study plan & SOPs of Aurigene Pharmaceutical Services Limited.

5.2 Good Laboratory Practice (GLP) Compliance

The study will be conducted as a non-GLP study and the procedures will be carried out as per the current standard operating procedures of the test facility and Study Plan. This study is not intended for regulatory submission.

5.3 Institutional Animal Ethics Committee (IAEC) Approval

This study will be performed in conformity with the Animal Welfare Law under the Animal License No. 3/PO/RcBi-S/RcBi-L/1999/CPCSEA and as per the approval of IAEC protocol No: IAEC/AUR/2022/11/44/PH/CIA/PV/87/R/F-106.

6. SAFETY PRECAUTIONS

Gloves, cap, face mask, gowns or lab coats and/or other appropriate personnel protective equipment will be used to handle the test and /or standard compound to ensure adequate personal health and safety and to avoid inhalation and skin contact with the test item. In case of eye contact, the eyes must be washed thoroughly with water and medical treatment to be sought. In case of skin contact, skin must be washed with soap and water with subsequent medical aid. Leak and spillage procedures: area of spillage will be ventilated; appropriate personnel protective equipment (gloves, cap, face mask, gowns and safety glasses) will be worn before entering the area. Area including floor and walls are flushed with water, wet mop or vacuum pump will be used to pick up spilled product. Unnecessary and unprotected personnel are kept away from entering in the spilled area.

7. PROPOSED STUDY SCHEDULE

Acclimatization	11 Jan 2023
Randomization	13 Jan 2023
Day of Sensitization	18 Jan 2023
Day of Booster injection	25 Jan 2023
Experimental Completion Date	15 Feb 2023

8. MATERIALS AND METHODS

8.1 Test & Reference item details

Name	:	[redacted]
Batch Number	:	[redacted]
Purity HPLC,	:	98.6%
Manufacture Date	:	07-Aug-2022
Retest Date	:	06-Feb-2023
Correction Factor	:	1.014

Storage Condition	:	2-8°C

Name	:	[redacted]
Batch Number	:	[redacted]
Purity HPLC,	:	[redacted]
Manufacture Date	:	30-Nov-2022
Retest Date	:	NA
Correction Factor	:	1.009
Storage Condition	:	Toom temperature

Name	:	Tofacitinib Citrate
Make	:	A CHEM BLOCK
Lot Number	:	19845A
Catalog no.		G-6156
Correction Factor		1.61
Storage Condition	:	0-8°C

Name	:	Dexamethasone
Make	:	Sigma
Lot Number	:	BCBV3214
Molecular weight		392.46106 g/mol
Storage Condition	:	2-8°C

8.2 Materials

Name	Make	Catalog no.	Batch/ Lot no.
Bovine Type II Collagen	Chondrex Inc	20021	220354
Incomplete Freund's adjuvant (IFA)	Sigma Aldrich	F5506	SLCM8843
SBE-β-CD	CycloLab R&D Ltd.	NA	47K441120
Acetic Acid	Sigma Aldrich	695092	STBK2788

Name	Make	Catalog no.	Batch/ Lot no.
CD3 Monoclonal Antibody (eBioG4.18 (G4.18)), FITC	eBioscience	11-0030-82	TBU
CD4 Monoclonal Antibody (OX35), Super Bright™ 436	eBioscience	62-0040-82	TBU
FOXP3 Monoclonal Antibody (FJK-16s), APC	eBioscience	17-5773-82	TBU
CD152 (CTLA-4) Monoclonal Antibody (WKH203), PE	eBioscience	12-1520-82	TBU
CD278 (ICOS) Monoclonal Antibody (C398.4A), PE-Cyanine5	eBioscience	15-9949-82	TBU
eBioscience™ Fixable Viability Dye eFluor™ 780	eBioscience	65-0865-14	TBU
Foxp3 / Transcription Factor Staining Buffer Set	eBioscience	00-5523-00	TBU
Stain buffer	BD	554656	2063415
Cell strainer	Mitenyi	130-098-462	5220420004
Cytoflex Daily QC beads	Beckman Coulter	B53230	12APA
Sheath Fluid	Beckman Coulter	B51503	6410099
DPBS Media	Sigma	59321C	SLCJ2830
Purified Mouse Anti-Rat CD32 (Fc Block)	BD	550271	TBU
FBS	Gibco	30-2020	80219215
DMSO	Sigma	D2650	RNBK7771
Red Blood Cell Lysing Buffer Hybri-Max	Sigma	R7757	SLCJ2830
MEM non-essential amino acids	Gibco	11140050	2517549
2-Mercaptoethanol (14.3 M)	Sigma	M3148	BCBV8182
Pen strep	Gibco	15140-122	2441841
Methyl cellulose	Sigma	M0262	SLCF9694
Tween 80	Sigma	P4780	K017022

8.3 Test System Details

Species and strain	:	SD Rats
Breeder/ Supplier	:	In-house

Animals will be acclimatized to study room of minimum 3-4 days. On day -12, all animal's body weight will be taken. Based on the body weight animals will be grouped into respective groups n=8 as per below mentioned table.

Group No.	Treatment	Dose	Dose volume (mL/kg)	Route and Frequency of Administration	Animal number
G1	Vehicle Control	Vehicle 5ml/kg	5	S c & q.d	1-8
G2	Disease control+ Vehicle	Vehicle 5ml/kg	5	S c & q.d	9-16
G3	Disease control+ Dexamethasone	0.1 mg/kg	5	p.o & q.d	17-24
G4	Disease control+ [redacted]	0.15 mg/kg	5	S c & q.d	25-32
G5	Disease control+ [redacted]	2.5 mg/kg	5	S c & q.d	33-40
G6	Tofacitinib	30 mg/kg	5	p.o & q.d	41-48

9.2 Experimental Procedure:

All animals will be administered with 500µg of Collagen IFA emulsion at base of the tail on day -7 except G1. On day 0 G1 & G2 animals will be injected with the vehicle through subcutaneous route. G3 and G6 animals will be administered with the dexamethasone at 0.1mg/kg and tofacitinib at 30 mg/kg through oral route respectively. G4 & G5 animals were injected with the [redacted] at 0.15 mg/kg & [redacted] at 2.5 mg/kg through the subcutaneous route. After dosing on day 0, paw volume will be measured using micrometer or plethysmometer and booster dose 100µg of collagen emulsion (IFA) will be administered at the base of tail. Body weight and arthritic clinical scores of all animal will be measured and recorded every day after booster dose. Paw volume will be measured weekly twice after booster dose. On day 21 i.e termination day, all animals body weight and clinical score measurements will be recorded. After dosing of the vehicle and test formulation blood collection will be done at different time points for pharmacokinetics. For blood collection sparse sampling will be done in 5 animals from each group at 0, 15, 60,120, 240, 360, and 480 minutes and each time point n=3. After blood collection, animals will be euthanized, both hind paws will be collected for histopathology analysis. Another 3 animals will be processed for spleen collection in C tubes containing 2 mL of wash media. Detail procedure for single cell suspension described in 9.3 section. Spleen cells will be stained with different antibodies for FACS analysis or frozen for later staining and FACS analysis

9.2.1 Preparation of Vehicle & Test Compound:

9.2.1.1 Preparation of 1% w/v Tween 80 Stock Solution:

1 gram of Tween 80 will be weighed in a glass container and dissolved in purified water (example 50 mL) by stirring over magnetic stirrer. The contents will be transferred, and final volume will be made up 100 mL with water in a suitable measuring cylinder to get strength of 1% Tween 80 stock solution. The prepared solution will be stored in a suitable glass bottle.

9.2.1.2 Preparation of 1% w/v Methyl cellulose Stock Solution:

1 gram of methyl cellulose will be weighed in a glass container and dissolved in purified water (example 50 mL) by stirring over magnetic stirrer. The contents will be transferred, and final volume will be made up 100 mL with water in a suitable measuring cylinder to get strength of 1% methyl cellulose stock solution. The prepared solution will be stored in a suitable glass bottle.

9.2.1.3 Preparation of 0.01N Acetic acid:

Preparation of 1N acetic acid solution: 0.57ml of glacial acetic acid was dissolved in 10ml of water.

Stock Concentration (X_1)	Volume of stock added to media (mL)	Volume of media taken (mL)	(Conc X_2)	Dilution
Glacial Acetic acid 1N	1	9	0.1N	1:10
Glacial Acetic acid 0.1N	3	27	0.01N	1:10

9.2.1.4 Preparation of Dexamethasone 0.1mg/kg:

Preparation of 0.1mg/mL Stock: Approximately weigh ~1mg of Dexamethasone (including salt correction) and transfer into a clean pestle. Add 1 mL of 1%w/v Tween 80 to it slowly while grinding it uniformly using a pestle; add remaining 4 mL of 1%w/v Tween 80. Later add 5 mL of 1% w/v methyl cellulose to make up the total volume to 10 mL.

Preparation of 0.1mg/kg: 5 mL of 0.1mg/mL stock will be diluted with vehicle make up to volume 25mL.

9.2.1.5 Preparation of Tofacitinib 30mg/kg: 154.56 mg of Tofacitinib citrate (Including salt correction) will be weighed and transferred into a clean Mortar pestle. 8 mL of 1% Tween 80 will be added dropwise and the compound will be triturated using a pestle to form a fine suspension. Then 8 mL of 1% MC was added dropwise.

9.2.1.6 Preparation of [redacted] (2.5mg/kg):

25 mg of [redacted] will be weighed and transferred in appropriate flask. 1.5 gm of Sulfobutyl Ether Beta Cyclo- Dextrin Sodium (SBECD) will be added in the flask, finally volume will be made upto the 50mL with the water. Flask will be sonicated till the solution becomes clear. the resultant solution will be filtered through the 0.22 µm filter. Same formulation will be used for the dosing up to three days by keeping the formulation in the 2-8°C.

9.2.1.7 Preparation of [redacted] (0.15mg/kg):

1.5 mg of [redacted] will be weighed and transferred in appropriate flask. 120 mg of Sulfobutyl Ether Beta Cyclo- Dextrin Sodium (SBECD) will be added in the flask, finally volume will be made upto the 50mL with the water. Flask will be sonicated till the solution becomes clear. the resultant solution will be filtered through the 0.22 µm filter. Same formulation will be used for the dosing up to three days by keeping the formulation in the 2-8°C.

9.2.1.8 Preparation of Sensitization Emulsion

Preparation of Collagen stock 4mg/mL stock:

Three bottles of collagen (30 mg) will be transferred to amber color bottle with the help of forceps, 6 mL of 0.01N acetic acid will be added, and solution will be kept under overnight stirring at 4°C. 0.5 mL of 0.01N acetic acid will be added to each collagen vial to remove any leftover collagen which adheres to bottle and vials will be keeping as such at 4°C. Next day all collagen solutions will be mixed before preparation of emulsion.

Preparation of Emulsion:

Collagen IFA Emulsion:

7.3 mL of oil phase (IFA) will be transferred into glass syringe having 3-way connector stoppers. 7.3 mL of Pre-mix aqueous phase will be added drop by drop in glass syringe while homogenizing using homogenizer (Make-REMI Motors, Model- RQ127) on ice. After complete addition of aqueous phase emulsion will be homogenized for 2- 3 mins for ensuring complete mixing.

Group 2 to Group 6 animals will be injected of 500µg of Collagen by injecting total emulsion volume of 250µl intradermally into the base of tail distributed at 2 sites.

9.2.1.9 Preparation of booster emulsion on day 0:

Preparation of Collagen 4 mg/mL stock: one bottle of collagen (10 mg) will be transferred to amber color bottle with the help of forceps, 2 mL of 0.01N acetic acid will be added, and solution will be kept under overnight stirring at 4°C. 0.5 mL of 0.01N acetic acid will be added to vial to remove any leftover collagen, which adheres to bottle, and vial will be keeping as such at 4°C.

Preparation of Emulsion:

Collagen IFA Emulsion(100µg): 10 mL of oil phase will be transferred into glass syringe having 3-way connector stoppers. 2 mL of Pre-mix aqueous phase will be added drop by drop in glass syringe while homogenizing followed by addition of 8 mL of 0.01N acetic acid drop by drop in glass syringe while homogenizing. After complete addition of aqueous phase emulsion will be homogenized for 2- 3 mins for ensuring complete mixing

Paw volume will be measured by using Micrometer for all animals before booster dose injection. All animals will receive 100µg of Collagen by injecting total emulsion volume of 250µl intradermally into the base of tail at 2 sites.

9.2.2 Preparation of media & Reagents:

9.2.2.1 Preparation of 55 mM 2-mercaptoethanol: 7.7 µL of 2-mercaptoethanol (14.3 M) will be added to 2.0 mL of RPMI-1640 medium to get a concentration of 55 mM 2-mercaptoethanol

9.2.2.2 Preparation of wash media for splenocytes collection: 1000 mL of wash media will be prepared aseptically using 979 mL of RPMI-1640, 20 mL of FBS and 1 mL of 55 mM 2-mercaptoethanol

9.2.2.3 Preparation of tissue culture media for splenocytes stimulation: 500 mL tissue culture media will be prepared aseptically using 440 mL of RPMI-1640, 50 mL of FBS, 5 mL of MEM non-essential amino acid, 5 mL of pen strep solution and 0.5 mL of 55 mM 2-mercaptoethanol.

9.2.2.4 Preparation of Foxp3 buffer: Dilute 1 part of Fixation/Permeabilization concentrate with 3 parts of Fixation/Permeabilization diluent prior to use. 30 mL of concentrate will be added to 90 mL diluent.

9.2.2.5 Preparation of blocking antibody: Blocking antibody will be prepared using 1:100 dilution of anti-rat Fc Block antibody CD32 in Stain buffer. 120µL of anti-Rat Fc Block antibody CD32 will be made up to 12 mL with stain buffer.

9.2.2.6 Preparation of CD3, CD4, ICOS antibodies: 40µl of CD3 antibody, 80µl of CD4 and 12µl of ICOS antibody will be mixed with 4 ml of stain buffer. 100µl will be added each well.

9.2.2.7 Preparation of Foxp3 & CTLA4 antibodies: 200µl of Foxp3 antibody and 200µl of CTLA4 antibody will be mixed with 4 ml of stain buffer. 100µl will be added each well.

9.2.2.8 Preparation of Fixable Viability Dye: FVD are supplied as a pre-diluted solution prepared in high-quality, anhydrous DMSO. 3µl aliquots of pre-diluted FVD solution will be prepared and stored at -70°C.working solution of the FVD will be prepared by diluting it 1:1000, in stain buffer. 100 µL/well will be added. For example, for 60 wells, add 6 µL of FVD to 6 mL of stain buffer.

FMO Control and Samples staining buffer composition		
Control/ sample name	**N=**	**Composition**
Unstained Spleen control	2	100 µL Stain buffers
CD3 FMO Control	2	100 µL all antibody stain except CD3
CD4 FMO Control	2	100 µL all antibody stain except CD4
ICOS FMO Control	2	100 µL all antibody stain except ICOS
Foxp3 FMO Control	2	100 µL all antibody stain except FOXP3
CTLA4 FMO Control	2	100 µL all antibody stain except CTLA4
Each group	3	100 µL all antibody CD3, CD4 and ICOS followed by Fixable Viability Dye, followed by Foxp3 and CTLA4

Table no. 9.1: FMO Control and Samples staining buffer composition

Note: All the above antibodies will be prepared in stain buffer except Foxp3 and CTLA4 antibodies will be prepared in perm buffer

9.3 Preparation of single cell splenocytes suspension:

9.3.1 Animals will be sacrificed by cervical dislocation and spleen will be collected in 15 mL centrifuge tube containing 5 mL wash media.

9.3.2 All the tubes will be brought into laminar air flow unit and processed as mentioned below.

9.3.3 Spleen collected will be placed C tubes containing 2 mL of wash media.

9.3.4 Single cell suspension will be prepared by using gentle MACS instrument using predefined protocol m-spleen-01-01 and will be run for 2-3 times, after suspension will be passed through 70micron cell strainer.

9.3.5 Using the 1 mL syringe plunger, press the tissue through the strainer to strain the cell suspension.

9.3.6 Carefully rinse the cell strainer into each tube with ~10 mL of wash media. Keep cell suspension cold on ice.

9.3.7 The spleen cell suspension will be spin down for 10 minutes at approximately 400g (~1400 rpm). Supernatant will be discarded, and pellet will be re-suspended in 1 mL of cold red blood cell lysing buffer and will be kept on ice for ~5 mins.

9.3.8 ~10 mL of cold wash media will be added to the tube and spin down for 10 minutes at approximately 400 g, discard the supernatant.

9.3.9 Obtained pellet will be re-suspended in ~5-10 mL of tissue culture media and cells will be counted using haemocytometer or cell counter.

9.3.10 Cell number will be adjusted to ~10 x10^6/mL with splenocytes culture media and keep on ice until ready to plate into 96 wells plate. 100 µL of cell suspension will be added to each well to obtain 1x10^6 cells per well.

Note: Based on email communication with sponsor, if all antibodies are arrived at test facility at the end of the termination, FACS analysis will be carried at test facility. Details are captured in 9.4 section onwards. If not, single spleen cell suspension will be frozen in 20% DMSO FBS solution as described below and will be analysed after all the antibodies received.

9.3.11 Cells concentration will be adjusted to 20- 40 x 10^6 ml in FBS and keep in the ice of 4°C until freezing. DMSO 20% in FBS will be prepared. 0.5 ml of DMOS 20% will be added in the cryovial. 0.5 ml of the cells will be added on top of DMSO/FPBS and mix.

9.3.12 Vial will be kept in Isopropanol Freezing Container for 24 h in -80°C, then will be transferred to liquid nitrogen until use.

9.4 Blocking: After single cell suspension seeding into 96 well plate. Splenocytes will be blocked with 100µl/well anti-rat antibody CD32 Fc Block for 15 min at 4°C.

9.5 Staining:

9.5.4 After blocking, 100µl mixture of CD3, CD4 and ICOS antibodies will be added and incubated for 30 min at 4°C. After incubation cells will be washed 3 times with PBS.

9.5.5 Fixable Viability Dye will be added 100µl into the each well (1/1000) and incubated for 15 min at 4°C.

9.5.6 After incubation cells will be washed for 3 times with PBS.

9.5.7 Fixation/Permeabilization buffer will be added each well and incubate for 30 min at 4°C.

9.5.8 Plate will spin down then wash with perm buffer.

9.5.9 Foxp3 and CTLA4 antibodies mixture in perm buffer will be added and incubate for 30 min at 4°C.

9.5.10 Wash twice with perm buffer then wash once with staining buffer and fix until flow cytometry acquisition.

10. OBSERVATIONS

10.1 Mortality/Morbidity

Animals will be observed for mortality/morbidity once daily during the study.

10.2 Paw volume measurement

Animal paw volume will be measured and recorded twice weekly entire study duration using micrometer or Plethysmometer.

10.3 Arthritic score

All animals will be observed for arthritic symptoms. Each fore & hind limb will be scored based on signs, which will be scored between 0-4 and each rat will be scored both the fore & hind limb, so total score of 0-16 (sum of two fore limbs score & sum of two hind limbs score). Arthritic score will be recorded on day 0 onwards as throughout the study

Clinical Score Criteria

Score	Observations
0	No evidence of erythema or swelling
1	Erythema and mild swelling confined to mid foot or ankle joint
2	Erythema and moderate swelling extending from ankle to mid foot
3	Erythema and moderate swelling extending from ankle to metatarsal joints
4	Erythema and severe swelling ankle, foot and digits

10.4 Body Weight

Individual body weights will be recorded every day before the administration of test item throughout the experimental period.

10.5 Blood Collection

On day 21, all animals body weight and clinical score measurements will be recorded. For blood collection first five animals from each group will be used for the sparse sampling. Under isoflurane anaesthesia, blood sample (~0.3mL) will be collected from Tail vein in to 10% w/v K2 EDTA tubes (2 mg/mL of blood) at pre-defined time points (0.25, 1, 2, 4, 6, and 8hr). Blood samples will be centrifuged at 6000 rpm for 5 min at below 10°C and plasma will be separated and stored at -60°C to -80°C.

Table for blood collection

Group No	**D20 blood collection Time Points**						
	0h (±2min)	**0.25h (±5min)**	**1h (±5min)**	**2h (±5min)**	**4h (±5min)**	**6h (±5min)**	**8h (±5min)**
G1	1,2,3	1,4,5	2,3,4	1,2,5	3,4,5	1,2,3	1,4,5
G2	9,10,11	9,12,13	10,11,12	9,10,13	11,12,13	9,10,11	9,12,13
G3	17,18,19	17,20,21	18,19,20	17,18,21	19,20,21	17,18,19	17,20,21
G4	25,26,27	25,28,29	26,27,28	25,26,29	27,28,29	25,26,27	25,28,29
G5	33,34,35	33,36,37	34,35,36	33,34,37	35,36,37	33,34,35	33,36,37
G6	41,42,43	41,44,45	42,43,44	41,42,45	43,44,45	41,42,43	41,44,45

10.6 Study termination:

On day 21, all animals will be euthanized by Cervical dislocation, Spleen will be collected from the remaining three animals from each group. in addition, both hind limbs and knee joints will be collected and stored in 4% formaldehyde, decalcified and embedded in paraffin to conduct histopathology

Table for Spleen collection:

Group No.	**Animal Number**
G1	6,7,8

G2	14,15,16
G3	22,23,24
G4	30,31,32
G5	38,39,40
G6	46,47,48

10.7 Histopathology:

Histological processing will be performed on cross sections of knee joint for each animal. Section will be taken from each block. Section will be stained for H&E. Analysis of slides will be done for various parameters Knee and Ankle Inflammation, Ankle Pannus, Knee Pannus, Ankle Cartilage Damage (Emphasis on small tarsals), Knee Cartilage Damage, Ankle Bone Resorption, Knee Bone Resorption and Periarticular Matrix Deposition.

11. DATA PRESENTATION AND REPORTS

Upon completion of study, electronic copy of data in the form of power point presentation will be shared along with raw data (MS EXCEL) for sponsor review. If required, based on sponsor request draft report will be prepared and shared with sponsor for review. After finalization of report by sponsor, signed copy of original report will be send to sponsor. All raw data (or authenticated copies thereof) and a copy of the final report, original study plan and amendments will be retained in the archive for a period of 2 years, by then the sponsor needs to indicate further storage at additional cost.

12. ANIMAL ETHICS

Animals considered to be in pain, distress or moribund will be examined by authorized personnel (qualified technicians/ study director or deputy/ experienced veterinarian) and, if deemed necessary, humanely sacrificed to minimize undue pain or suffering. Therefore, animals in extremis are killed by a painless method to comply with the law on the protection of animals.

13. REFERENCES

1) Seeuws S et al., "A multiparameter approach to monitor disease activity in collagen-induced arthritis". Arthritis Res Ther. 2010, 12:R160.
2) Min Shi et al., "The protective effects of chronic intermittent hypobaric hypoxia pretreatment against collagen-induced arthritis in rats". Journal of Inflammation (2015) 12:23

3) Xue X et al., "Pharmacologic modulation of RORγt translates to efficacy in preclinical and translational models of psoriasis and inflammatory arthritis". Scientific Reports | 6:37977 | DOI: 10.1038/srep37977

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1. LIST OF ABBREVIATIONS

Abbreviation		Full Form/ Description
~	:	Approximately
%	:	Percentage
°C	:	Degree Centigrade
µL	:	Microlitre
µm	:	Micrometer
Ani# or A# or A.No.	:	Animal number
ANOVA	:	Analysis of Variance
APSL	:	Aurigene Pharmaceutical Services Ltd.
CPCSEA	:	Committee for the Purpose of Control and Supervision of Experiments on Animals
FBS	:	Fetal Bovine Serum
FMO	:	Fluorescence Minus One
GLP	:	Good Laboratory Practice
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
i.m.	:	Intramuscular
IAEC	:	Institutional Animal Ethics Committee
IVC	:	Individually ventilated cages
K_2EDTA	:	Dipotassium Ethylene Diamine Tetra-acetic Acid
kb	:	Kilobase
M. Pharm.	:	Master of Pharmacy
M.V. Sc.	:	Master of Veterinary Science
mg	:	Milligram
mL	:	Millilitre
NA	:	Not applicable
nm	:	Nanometre
PH	:	Pharmacology
RBC	:	Red Blood Cells
rpm	:	Revolutions Per Minute
SEM	:	Standard Error of Mean
SOP	:	Standard Operating Procedure
Sp	:	Splenocytes
TBU	:	To be updated

2. STUDY & STUDY PERSONNEL DETAILS

Study No.	:	[redacted]
Study Title	:	Evaluation of [redacted] vaccine and [redacted] vaccine test formulations for tolerability and immunogenicity in BALB/c mice
Study Director	:	Prashant Yallappa Vadnal, M. Pharm. Pharmacology Department Aurigene Pharmaceutical Services Limited Bollaram Road, Miyapur Hyderabad-500, 049, Telangana, India Telephone No.: + 91-40-44657340 Email: prashantyallappa_v@aurigeneservices.com
Head Function	:	D. Ravi Krishna Babu, M. Pharm. Department of Pharmacology, Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad -500 049, Telangana, India Tel No.: +91-40-44657345 E-mail: ravikrishna_b@aurigeneservices.com
Sponsor	:	[redacted]
Responsible Scientists	:	Ch Sandeep Reddy, M. Pharm. Kishor Chitalkar, M. Pharm. Sandeep M, M. Pharm.
Animal Welfare Officer	:	Dr. Deeksha Hattimare, M.V. Sc.

3. OBJECTIVE

The objective of the study is to determine the tolerability as well as humoral and cell-mediated immune response against [redacted] vaccine and [redacted] vaccine test formulations in BALB/c mice using serum & splenocytes.

4. INTRODUCTION

The [redacted] are DNA viruses. [redacted] are ubiquitous infecting many different mammalian species including humans. There are five known human polyoma viruses. [redacted] virus are two polyoma viruses identified nearly three decades ago. Recently [redacted] and [redacted] viruses have been isolated from humans. The exact role of these three newly discovered viruses in human disease is not known. Most human [redacted] disease is caused by [redacted] viruses which are usually acquired in childhood. [redacted] reaches the brain and causes progressive multifocal leukoencephalopathy, a demyelinating disease of the central nervous system with a high mortality rate. [redacted] is urotheliotropic and its reactivation causes a form of interstitial nephritis, known as [redacted] virus associated nephropathy which is associated with high graft loss if not recognized early.

The genus of [redacted] belongs to the [redacted]. [redacted] are icosahedral, non-enveloped double stranded circular super coiled viruses. They measure 40.5 – 44 nm in diameter and comprise 88% protein and 12% DNA. The viral DNA genome has around 5000 base pairs organized in three different functional regions: the early coding and late coding regions separated by a non-coding regulatory region. [redacted] is acquired in early childhood whereas [redacted] later, 3–4 years versus 10–14 years. The mode of transmission of these viruses is not well known. Transmission through the feco-oral and respiratory routes has been suggested. Nearly all healthy adults are asymptomatically infected with human polyomaviruses. In immunosuppressed individuals, the infection can reactivate and cause disease.

The outer capsid proteins of [redacted] are structurally similar to the capsids of [redacted] Based on success of the [redacted] vaccine technologies, [redacted] vaccines targeting [redacted] were developed. [redacted] is a small, non-enveloped deoxyribonucleic acid (DNA) virus that infects skin or mucosal cells. The circular, double-stranded viral genome is approximately 8 kb in length. The genome encodes for 6 early proteins responsible for virus replication and 2 late proteins, L1 and L2, which are the viral structural proteins. [redacted] vaccines are vaccines that prevent infection by certain types of human papillomavirus. The [redacted] vaccines require two or three doses depending

on a person's age and immune status. The [redacted] vaccines provide protection for at least 5 to 10 years.

In this study, tolerability as well as humoral and cell mediated immune response against [redacted] vaccine and [redacted] vaccine test formulations will be evaluated in the BALB/c female mice, by measuring antibodies in serum and cell surface marker analysis/cytokine secretion in *in vitro* stimulated splenocytes at different time points and dose levels.

5. REGULATORY COMPLIANCE AND APPROVALS

5.1Test guidelines

This study is not based on any specific test guidelines. However, it will be conducted in compliance with the mutually agreed study plan & SOPs of Aurigene Pharmaceutical Services Limited.

5.2 Good Laboratory Practice (GLP) compliance

The study will be conducted as a non-GLP study and the procedures will be carried out as per the current Standard Operating Procedures (SOP) of the test facility and mutually agreed Study Plan. This study is not intended for regulatory submission.

5.3 Institutional Animal Ethics Committee (IAEC) Approval

This study will be performed in conformity with the Animal Welfare Law under the Animal License No. 3/PO/RcBi-SL/99/CPCSEA and as per the approval of IAEC protocol No: IAEC/AUR/2023/03/47/PH/vaccine/PV/150/M/F-70, on 20th March 2023.

6. SAFETY PRECAUTIONS

Gloves, cap, face mask, gowns or lab coats and/or other appropriate personnel protective equipment will be used to handle the test and /or standard compound to ensure adequate personal health and safety and to avoid inhalation and skin contact with the test item. In case of eye contact, the eyes must be washed thoroughly with water and medical treatment to be sought. In case of skin contact, skin must be washed with soap and water with subsequent medical aid. Leak and spillage procedures: area of spillage will be ventilated; appropriate personnel protective equipment (gloves, cap, face mask, gowns and safety glasses) will be worn before entering the area. Area including floor and walls will be flushed with water, wet mop or vacuum pump will be used to pick up spilled product. Unnecessary and unprotected personnel will be kept away from entering in the spilled area.

7. PROPOSED STUDY SCHEDULE (Tentative)

Animal Procurement	24th April 2023
Acclimatization	2nd May 2023
Randomization	5th May 2023
Blood collection	8th May 2023, 29th May 2023, 19th June 2023
Test formulation administration	9th May 2023, 30th May 2023, 20th June 2023
In life completion date (Final blood and Splenocytes collection)	4th July 2023 and 5th July 2023
Splenocytes staining and acquisition	6th July2023 and 7th July 2023 (Staining), 7th July 2023 and 8th July 2023 (Acquisition)

8. MATERIALS AND METHODS

8.1 Test Item Information

Test item Name: [redacted] Mfg. by: [redacted] Batch no: [redacted] Date of Mfg.: Jan 2023 Expiry date: Dec 2024 Purity if any: NA	**Test item Name:** [redacted] Mfg. by: [redacted] Batch no: [redacted] Date of Mfg.: Mar 2023 Expiry date: Feb 2025 Purity if any: NA
Test item Name: [redacted] Mfg. by: [redacted] Batch no: [redacted] Date of Mfg.: Mar 2023 Expiry date: Feb 2025 Purity if any: NA	**Test item Name:** [redacted] Mfg. by: [redacted] Batch no: P[redacted] Date of Mfg.: Mar 2023 Expiry date: Feb 2025 Purity if any: NA
Test item Name: [redacted] Mfg. by: [redacted] Batch no: [redacted] Date of Mfg.: Mar 2023 Expiry date: Feb 2025 Purity if any: NA	**Test item Name:** [redacted] Mfg. by: [redacted] Batch no: [redacted] Date of Mfg.: Mar 2023 Expiry date: Feb 2025 Purity if any: NA

8.2 Materials and Equipment Used

Reagents and Materials	Make	Catalog No.	Batch no./ Lot no.	Expiry date
FBS	Gibco	10270-106	2384436	30 June 2026
RPMI 1640 Media	Gibco	A10491	TBU	TBU

Reagents and Materials	Make	Catalog No.	Batch no./ Lot no.	Expiry date
Countess chamber slides	Thermo Fisher	C10283	I30B12231	NA
HPLC grade Water	Rankem	W0020	TBU	TBU
Red Blood Cell Lysing Buffer Hybri-Max	Sigma	R7757	RNBK5280	31 Oct 2023
MACS SmartStrainers (70 µm)	Miltenyi	130098462	5210801920	05 Aug 2023
Gentle MACS C tubes	Miltenyi	130096334	5201115010	15 Nov 2023
2-Mercaptoethanol (14.3 M)	Sigma	M3148	TBU	TBU
MEM non-essential amino acids	Gibco	11140050	TBU	TBU
96-well cell culture plates with lid, round well, treated, sterile	Corning	3795	07321004	31 Mar 2024
CytoFlex Daily QC beads	Beckman Coulter	B53230	02ANAF	19 Feb 2024
Sheath Solution	Beckman Coulter	B51503	6410072	20 Oct 2023
Stain Buffer	BD	554656	9329560	27 Sep 2023
Fixation buffer	BD	554655	TBU	TBU
CryoStor cell cryopreservation media	Sigma	C2874	MKCQ7025	31 Jan 2024
Brilliant Stain Buffer Plus	BD	566385	1277645	31 Oct 2024
BV421 Rat Anti-Mouse CD3	BD	564008	TBU	TBU
PE-Cy™7 Rat Anti-Mouse CD4	BD	552775	TBU	TBU
FITC Rat Anti-Mouse CD8a	BD	553030	TBU	TBU
PE Rat Anti-Mouse CD19	BD	557399	TBU	TBU
BV650 Rat Anti-Mouse CD45RA	BD	752984	TBU	TBU
CD45RO AF647 Antibody	Santacruz Biotechnology	sc-1183 AF647	TBU	TBU
BV650 Rat Anti-CD11b	BD	563402	TBU	TBU
APC Rat Anti-Mouse CD14	BD	560634	TBU	TBU
BV421 Rat Anti-Mouse CD86	BD	564198	TBU	TBU
PE anti-mouse CD163	BioLegend	156704	TBU	TBU
FITC Rat anti-Mouse Ly-6G	BD	561105	TBU	TBU
Mouse Fc Block CD16/CD32	BD	553142	TBU	TBU
Zombi Aqua Fixable Viability Stain 510 kit	Bio legend	423101	TBU	TBU
Anti-Rat and Anti-Hamster Ig κ /Negative Control Compensation Particles Set	BD	552845	TBU	TBU
Perm/Wash Buffer	BD	51-2091KZ	TBU	TBU
Fixation/ Permeabilization solution	BD	51-2090KZ	TBU	TBU

Equipment	Make	Model
Biosafety cabinet	ESCO Life Sciences	AC2-6S8
Laminar Flow clean air workstation	Klenzaids Bioclean	1594
Tissue Dissociator	Miltenyi Biotec	GentleMACS Octo Dissociator
Refrigerated Centrifuge	Eppendorf	5417R
Refrigerated Centrifuge	Eppendorf	5810R
Centrifuge	Eppendorf	5810
-86°C Ultra Freezer	Thermo Forma	984
Cell Counter	Invitrogen	Countess 3FL
Flow cytometer	Beckman Coulter	Cytoflex S B75442
Biosafety cabinet 1300 series A2	Thermo	1389
Precision balance	Sartorius	BSA6202S-CW
IVC system	Citizen	V3
Anaesthesia machine	Midmark Matrx	Matrx VML
Refrigerator	LG	GRS352QLC
CO_2 incubator	New Brunswick	Galaxy 170S

8.3 Test System Details

Species and strain	:	BALB/c Mice
Breeder/ Supplier	:	Hylasco Biotechnology, Hyderabad
Number of animals in study	:	60 (10 animal per group; 6 groups)
Initial Age	:	5-7 weeks (at the time of acclimatization)
Sex	:	Female
Animal housing	:	Animals will be housed in group of 5 animals per cage in polypropylene cages containing irradiated corncob bedding material, with enrichment toys. Room temperature and relative humidity ranges of 22 ± 3°C and 50 ± 20%, respectively. Twelve-hour light/dark cycle provided via automatic timer. Ventilation will produce 10-15 air changes/ hour.
Feed and water	:	Animals will be fed with the Gamma Irradiated commercial rodent maintenance diet 1324 (Altromin). Purified water will be provided *ad libitum* in polypropylene bottles.

vaccine test items will be administered through the i.m. route to the respective treatment groups. Under ~3% isoflurane anesthesia, ~250 µL/animal blood will be collected into 1.5 mL micro centrifuge tubes to separate serum on day 21, day 42 and ~500 µL/animal blood will be collected on terminal day 57 and 58. Serum will be separated under sterile conditions in 0.5 mL labelled sterile micro centrifuge tubes, will be stored immediately at -70 ± 10°C and will be shipped to Sponsor through dry ice at the end of the study.

Body weight of all animals will be measured and recorded twice a week. Clinical signs of animals will be observed twice a week and recorded. Study terminations will be carried out in two days (day 57-58) such that from each group, 5 animals will be terminated on day 57 and remaining 5 animals on day 58. On study termination day i.e., on day 57 and day 58, after blood collection, animals will be sacrificed, and spleens will be processed into single cell suspension for *in vitro* re-stimulation with corresponding test item formulation under sterile conditions. After 40-48 hours incubation, supernatants collected under sterile conditions will be stored in sterile 1.5 mL micro centrifuge tubes at -70 ± 10°C immediately and will be shipped to sponsor through dry ice at the earliest. The *in vitro* stimulated splenocytes will be stained for various cell surface markers for flow cytometry analysis as per procedure mentioned below.

Animal sacrifice will be done on two days as mentioned above. *In vitro* stimulation, staining procedures and acquisition by flow cytometer activities will be done independently on different days with procedure as mentioned below.

8.4.3 Preparation of Media for Splenocytes collection:

8.4.3.1 Preparation of 55 mM 2-mercaptoethanol:

Under sterile tissue culture conditions, 12 µL of 2-mercaptoethanol (14.3 M) will be added to 3.0 mL of RPMI-1640 medium to get a concentration of 55 mM 2-mercaptoethanol.

8.4.3.2 Preparation of wash media for splenocytes collection:

Under sterile tissue culture conditions, 1200 mL of wash media will be prepared aseptically using 1174.8 mL of RPMI-1640, 24 mL of FBS and 1.2 mL of 55 mM 2-mercaptoethanol.

8.4.3.3 Preparation of tissue culture media for splenocytes stimulation:

Under sterile tissue culture conditions, 600 mL tissue culture media will be prepared aseptically using 527.4 mL of RPMI-1640, 60 mL of FBS, 6 mL of MEM non-essential amino acid, 6 mL of Pen strep solution and 0.6 mL of 55 mM 2-mercaptoethanol.

8.4.3.4 Preparation of blocking antibody: Blocking antibody will be prepared using 1:20 dilution of Mouse Fc Block antibody CD16/CD32 in Stain buffer. 720 µL Mouse Fc Block antibody CD16/CD32 will be added to 13680 µL stain buffer.

8.4.3.5 Preparation of antibody: Antibody cocktails 1 & 2 will be prepared for T cell & APC markers respectively as mentioned in below table. 1/200 dilution of each antibody will be mixed with Brilliant Stain buffer to obtain final antibody dilution of 1/400 in well.

FMO control for Antibody cocktail 1: 0.5 µL of each antibody will be made up to 100 µL with Brilliant Stain buffer.

FMO control for Antibody cocktail 2: 0.5 µL of each antibody will be made up to 100 µL with Brilliant Stain buffer.

Antibody cocktail 1: 18 µL of each antibody will be made up to 3600 µL with Brilliant Stain buffer.

Antibody cocktail 2: 18 µL of each antibody will be made up to 3600 µL with Brilliant Stain buffer.

a) Antibody cocktail 1 for T cell markers

T cell markers	Clone	Make	Catalog no.	Fluorescent dyes
CD3	553030	BD	564008	BV421
CD4	RM4.5	BD	552775	PE-Cy7
CD8	53-6.7	BD	553030	FITC
CD19	1D3	BD	557399	PE
CD45RA	14.8	BD	752984	BV650
CD45RO	UCH-L1	Santacruz Biotechnology	sc-1183 AF47	AF647

b) Antibody cocktail 2 for APC markers

APC markers	Clone	Make	Catalog no.	Fluorescent dyes
CD11b	M1/70	BD	563402	BV650
CD14	rmC5-3	BD	560634	APC
CD86	GL1	BD	564198	BV421
CD163	S15049F	BioLegend	156704	PE
Ly-6G	1A8	BD	561105	FITC

8.5.1 Preparation of Single Cell Splenocytes Suspension

- Animals will be sacrificed by cervical dislocation and spleen will be collected in sterile 2 mL micro centrifuge tube containing 1 mL of sterile wash media under sterile conditions.
- All the tubes will be brought into biosafety cabinet and will be processed as mentioned below.
- For Test conditions, splenocyte samples will be kept independently and not pooled between animals. For FMOs and pooled controls, splenocyte samples will be pooled by collecting 0.2 mL cell suspension from each animal sample.
- Collected spleen will be placed in Gentle MACS C tubes containing 2 mL of wash media.
- Single cell suspension will be prepared by using gentle MACS instrument using predefined protocol m-spleen-01-01 and will run for 3 times, after that suspension will be passed through 70 µm cell strainer.
- Cell suspension will be strained through the strainer.
- Cell strainer will be carefully rinsed into each tube with ~10 mL of wash media. Cell suspension will be kept on ice.
- The spleen cell suspension will be spun down for 10 minutes at approximately 1400 rpm at 4°C. Supernatant will be discarded, and pellet will be re-suspended in 1 mL of cold red blood cell lysing buffer and will be kept on ice for ~5 minutes.
- 10 mL of cold wash media will be added to the tube and spun down for 10 minutes at approximately 1400 rpm at 4°C and supernatant will be discarded.
- Obtained pellet will be re-suspended in 10 mL of tissue culture media and cells will be counted using cell counter.
- Cell number will be adjusted to $10x10^6$ cells/mL (based on live cell count) with sterile tissue culture media and kept on ice until ready to be plated into 96 wells plate. 100 µL of cell suspension will be added to each well to obtain $1x10^6$ cells per well. 10 million cells (i.e. 1 mL splenocytes suspension) per animal will be cryopreserved using freezing media and will be stored at -70 ± 10°C.

8.5.2 *In Vitro* Stimulation with test formulations

- 100 µL of cell suspension will be aliquoted into microplate wells and covered with lid.
- Cells will be activated using 10 µL per well of respective test formulation except for unstimulated unstained and stained controls.

- Cells used for FMO controls will be activated using 10 µL per well of test formulation [redacted].
- Final volume per well will be made up to 200 µL with required volume of sterile tissue culture media.
- For the unstimulated control wells, 100 µL of sterile tissue culture media will be added instead of test formulation.
- Plate will be incubated at 37°C in 5% CO_2 for 40-48 hours and stained as per procedure described in section 8.5.3.

8.5.3 T Cell & APC Markers Estimation in Mouse Splenocytes Using Flow Cytometry

- After incubation, plate will be spun down at approximately 1400 rpm for 10 minutes at room temperature. Supernatants will be collected in 96 well plate, pooled duplicate supernatants will be collected in 1.5 mL labelled sterile micro centrifuge tubes under sterile conditions, will be stored at -70 ± 10°C and will be shipped to Sponsor through dry ice at the earliest.
- Cells will be washed with 200 µL wash media and gently re-suspended. Cells will be spun down to pellet at 450 g for 5 minutes at 4°C and supernatant will be removed.
- Cells will be fixed and permeabilized by adding 100 µL per well of Fixation/ Permeabilization solution for 20 minutes at 4°C and pelleted down. Cell aggregation will be avoided by thoroughly mixing cells suspension prior to the addition of the Fixation/ Permeabilization solution.
- Cells will be washed twice using 1X BD Perm/Wash buffer (200 µL/well) and pelleted.
- 50 µL staining buffer containing Fc block will be added to wells and incubated for 10 minutes at 2-8°C.
- 50 µL antibody cocktail solutions 1 or 2 will be added to above and incubated for 45 minutes at 2-8°C.
- FMO (fluorescence minus one) control will be set with all the antibodies but one (using pooled splenocytes).

8.5.4 Proposed Plate Layout

Antibody Panel 1 T/B Cell Markers												
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
A	FMO CD3 Ab cocktail 1	A# 1-1	A# 5-1	A# 9-1	A# 13-1	A# 17-1	A# 21-1	A# 25-1	A# 29-1	G3: Media-Sp Unstim	Unstim + stained pSpleen	Stim G1: Test-Sp
B	FMO CD4 Ab cocktail 1	A# 1-2	A# 5-2	A# 9-2	A# 13-2	A# 17-2	A# 21-2	A# 25-2	A# 29-2	G3: Media-Sp Unstim	Unstim + stained pSpleen	Stim G1: Test-Sp
C	FMO CD8 Ab cocktail 1	A# 2-1	A# 6-1	A# 10-1	A# 14-1	A# 18-1	A# 22-1	A# 26-1	A# 30-1	G4: Media-Sp Unstim	Unstim+ unstained pSpleen	Stim G2: Test-Sp
D	FMO CD19 Ab cocktail 1	A# 2-2	A# 6-2	A# 10-2	A# 14-2	A# 18-2	A# 22-2	A# 26-2	A# 30-2	G4: Media-Sp Unstim	Unstim+ unstained pSpleen	Stim G2: Test-Sp
E	FMO CD45RA Ab cocktail 1	A# 3-1	A# 7-1	A# 11-1	A# 15-1	A# 19-1	A# 23-1	A# 27-1	G1: Media-Sp Unstim	G5: Media-Sp Unstim	L/D Unstim pSpleen	Stim G3: Test-Sp
F	FMO CD45RO Ab cocktail 1	A# 3-2	A# 7-2	A# 11-2	A# 15-2	A# 19-2	A# 23-2	A# 27-2	G1: Media-Sp Unstim	G5: Media-Sp Unstim	L/D Unstim pSpleen	Stim G3: Test-Sp
G	Stim G6: Test-Sp	A# 4-1	A# 8-1	A# 12-1	A# 16-1	A# 20-1	A# 24-1	A# 28-1	G2: Media-Sp Unstim	G6: Media-Sp Unstim	Stim G5: Test-Sp	Stim G4: Test-Sp
H	Stim G6: Test-Sp	A# 4-2	A# 8-2	A# 12-2	A# 16-2	A# 20-2	A# 24-2	A# 28-2	G2: Media-Sp Unstim	G6: Media-Sp Unstim	Stim G5: Test-Sp	Stim G4: Test-Sp

Antibody Panel 2 APC Markers												
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
A	FMO CD11b Ab cocktail 2	A# 1-1	A# 5-1	A# 9-1	A# 13-1	A# 17-1	A# 21-1	A# 25-1	A# 29-1	G3: Media-Sp Unstim	Unstim + stained pSpleen	Stim G1: Test-Sp
B	FMO CD14 Ab cocktail 2	A# 1-2	A# 5-2	A# 9-2	A# 13-2	A# 17-2	A# 21-2	A# 25-2	A# 29-2	G3: Media-Sp Unstim	Unstim + stained pSpleen	Stim G1: Test-Sp
C	FMO CD86 Ab cocktail 2	A# 2-1	A# 6-1	A# 10-1	A# 14-1	A# 18-1	A# 22-1	A# 26-1	A# 30-1	G4: Media-Sp Unstim	Unstim+ unstained pSpleen	Stim G2: Test-Sp
D	FMO CD163 Ab cocktail 2	A# 2-2	A# 6-2	A# 10-2	A# 14-2	A# 18-2	A# 22-2	A# 26-2	A# 30-2	G4: Media-Sp Unstim	Unstim + unstained pSpleen	Stim G2: Test-Sp
E	FMO Ly-6G Ab cocktail 2	A# 3-1	A# 7-1	A# 11-1	A# 15-1	A# 19-1	A# 23-1	A# 27-1	G1: Media-Sp Unstim	G5: Media-Sp Unstim	L/D Unstim pSpleen	Stim G3: Test-Sp
F		A# 3-2	A# 7-2	A# 11-2	A# 15-2	A# 19-2	A# 23-2	A# 27-2	G1: Media-Sp Unstim	G5: Media-Sp Unstim	L/D Unstim pSpleen	Stim G3: Test-Sp
G	Stim G6: Test-Sp	A# 4-1	A# 8-1	A# 12-1	A# 16-1	A# 20-1	A# 24-1	A# 28-1	G2: Media-Sp Unstim	G6: Media-Sp Unstim	Stim G5: Test-Sp	Stim G4: Test-Sp

H	Stim G6: Test-Sp	A# 4-2	A# 8-2	A# 12-2	A# 16-2	A# 20-2	A# 24-2	A# 28-2	G2: Media-Sp Unstim	G6: Media-Sp Unstim	Stim G5: Test-Sp	Stim G4: Test-Sp
L/D= Live Dead stain; Unstim= Unstimulated; Stim= Stimulated; ctrl= Control; pSpleen = pooled Splenocytes; Sp= Splenocytes, FMO samples are stimulated with PCT/PVV/NP/002A/23 test formulation. Animal test samples are stimulated with test formulations of respective group.												

- Plates will be spun down at 450 g for 5 minutes at 4°C.
- Supernatant will be removed, and cells will be re-suspended in 200 µL 1X BD Perm/Wash buffer. Plates will be spun down at 450 g for 5 minutes at 4°C twice for two consecutive washes.
- Cells will be re-suspended in stain buffer, stored at 4°C overnight and next day, plate acquisition will be done by flow cytometer.
- Data will be acquired from ≥10000 gated events and analysed using CytExpert software.

9. OBSERVATIONS

9.1 Mortality/morbidity and clinical signs observation

Animals will be observed once daily, while clinical signs of animals will be observed twice a week and same will be captured in clinical signs observations record till day 57. Animals will be observed for injection site reaction ~24 hrs after each dose administration.

9.2 Body weight

Individual animal body weights will be recorded twice a week before the administration of test formulation throughout the experimental period.

9.3 Blood collection, serum separation & storage

On day 0, day 21, day 42 under ~3% isoflurane anaesthesia, blood samples (~250 µL/animal) will be collected in to 1.5 mL micro centrifuge tubes from all animals. Blood samples will be kept in slanting position for 30 to 45 minutes, then samples will be centrifuged at 6000 rpm for 10 minutes at 4°C and serum will be separated under sterile conditions, transferred in another sterile 0.5 mL micro centrifuge tube and stored at -70 ± 10°C immediately after collection.

Terminal blood sample collection will be carried two days in such a way that 5 animals from each group will be done on day 57 and remaining 5 animals on day 58. On day 57 and day 58, all animals body weights and clinical score observations will be recorded. The blood samples (~500 µL/animal) will be collected from retro orbital plexus into 1.5 mL micro centrifuge tubes from all animals. Utmost care will be taken to prevent haemolysis of blood at all bleeding time points. Serum will be collected in sterile 0.5 mL micro centrifuge tubes under sterile conditions as mentioned above and will be stored at -70 ± 10°C immediately after collection. Serum samples from all bleeding time points will be collectively shipped to Sponsor through dry ice for analysis.

9.4 Study termination

Study termination will be carried out on two days in such a way that 5 animals from each group will be sacrificed on day 57 and remaining 5 animals on day 58. On day 57-58, after blood collection, animals will be sacrificed by cervical dislocation and spleen will be collected in sterile 2 mL micro centrifuge tube containing 1 mL of sterile wash media under sterile conditions. All the tubes will be brought into biosafety cabinet in tissue culture lab and will be processed as per steps mentioned in section 8.5.1. Splenocyte samples will be kept independently and not pooled between animals.

10. DATA ANALYSIS

10.1 Statistical Analysis

The percent change in body weight with respect to day 0 will be calculated using below formula.

% Change in body weight = {(Body weight on the given day- 0-day body weight)/ 0-day body weight} *100

Graphs and statistical analysis will be performed using One way or Two-way ANOVA with GraphPad Prism 9 (Graph Pad Software). When applicable, results will be presented as a Mean ± SEM.

11. AMENDMENTS AND DEVIATIONS PROCEDURES

This study protocol may be amended or be subjected to alterations. In each case, any major amendment to the approved Study protocol and the reasons for such amendments along with potential impact towards the overall objective and conclusions of the study will be put in writing and realized only after written consent from the Study Director and Sponsor. Any major unintended change(s) occurring during the execution of the study will be reported as Study Protocol Deviation report. All amendments and deviations to the Study protocol will be documented and incorporated in final report. The study protocol, amendment and report which require signature will be shared via DocuSign to all the signatories.

12. DATA PRESENTATION AND REPORTS

Upon completion of study, electronic copy of data in the form of power point presentation will be shared along with raw data (MS EXCEL) for sponsor review. Subsequent to Flow cytometry analysis, draft study report will be prepared and shared with Sponsor for review. After finalization of report by Sponsor, signed copy of original report will be sent to Sponsor. As per